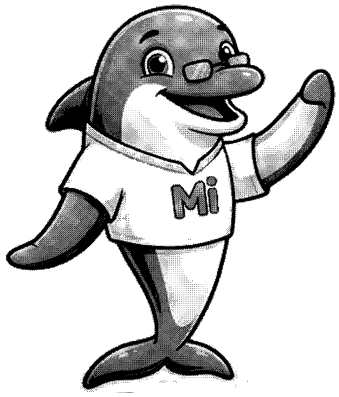
Math Inquirers

Extension Workbook

with answers

Level 6

Self-study | Ages 11-13



My name _____

About this book

Math Inquirers Extension Workbook, Level 6
Self-study practice with answers.

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Good to know

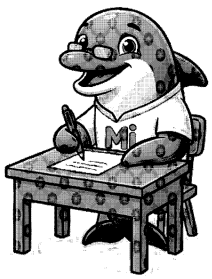
- This is a private-enrichment book. It is not an official school or examination book, and it makes no promise about any test result.
- We have checked the answers with care. If you find a mistake, please tell us so we can fix it.
- Printed copies are black and white. The soft copy has the same pages in colour.

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Welcome!

This book is for you to explore at your own pace. Every unit has 7 pages. You can start on your own. A grown-up can help if you like.

How to use this book

1. **Read the Learn it page first. It shows how the idea works.**
2. **Do the Start here pages. Use a pencil. Mistakes help you learn.**
3. **Stuck? Find the hint. It is printed upside down.**
4. **Do Check yourself. Check the answers. Wrong? Read the Why.**

For the grown-up

- Practice for the same ideas as the Holiday Program (our school-holiday course), with new numbers and stories. You do not need to have joined it: every unit starts with an example.
- 3 short sessions a week, 20 to 25 minutes each, plus the weekend page **Go further** (see the pace plan).
- This is a self-study book. Your child marks their own work: each answer at the back says the answer first, then a short reason (Why) in plain words, and the page of the question. A child who reads slowly may like a helper to read the Why with them. The answers start on page 79: please ask your child to try first.
- Check yourself score 2 or less? Read the Learn it page again and redo one Start here page.
- Beyond Level 6 is ahead of school. You may skip it.

The four page names

- ● **Start here.** The main path. Do these first.
- ◆ **Think harder.** Longer problems. Hints are printed upside down.
- ★ **Go further.** A weekend page. Big challenges, just for fun.
- ✓ **Check yourself.** Four quick questions at the end.

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3	**Below zero** Below zero · follows Holiday Program Lesson 3 · answers on page 84	page 20
4	**Algebra machines** Algebra machines · follows Holiday Program Lesson 4 · answers on page 86	page 27
5	**Every possible way** Every possible way · follows Holiday Program Lesson 5 · answers on page 88	page 34
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My pace plan

Numbers in the table are page numbers. One unit a week for 10 weeks. Three short sessions a week, about 20 to 25 minutes each. **Go further** is a weekend page: do it on a weekend, when there is time to think. Tick the box when the week is done.

Week	Unit	Session 1 Learn it, Start here	Session 2 Start here	Session 3 Think harder, Check yourself	Weekend page Go further	Done
1	**Code the pattern**	6-7	8-9	10 and 12	11	☐
2	**Prime hunters**	13-14	15-16	17 and 19	18	☐
3	**Below zero**	20-21	22-23	24 and 26	25	☐
4	**Algebra machines**	27-28	29-30	31 and 33	32	☐
5	**Every possible way**	34-35	36-37	38 and 40	39	☐
6	**Fair or unfair?**	41-42	43-44	45 and 47	46	☐
7	**Turtle geometry**	48-49	50-51	52 and 54	53	☐
8	**Secret codes**	55-56	57-58	59 and 61	60	☐
9	**Data detectives**	62-63	64-65	66 and 68	67	☐
10	**Showcase challenge**	69-70	71-72	73 and 75	74	☐
11	**Catch-up week**	Repeat any Check yourself where your score was 2 or less.				☐

Tips for the grown-up

- Short and often beats one long session. Stop while your child still wants more.
- Go further has big challenges. It is fine to skip it, or to do only one question. Missed a week? Carry on.

Learn it: Sequences and the nth term

My goal. I can find the rule for the nth term of a pattern and use it for the 100th term. I can test my rule with a loop.

A rule for the nth term jumps straight to any term, such as the 100th. The difference tells you which times table to compare with. A loop in code adds the difference again and again, so you can test your rule.

Worked example

A sequence begins 9, 14, 19, 24, ... Find the rule that gives term n. Then find the 100th term.

n	1	2	3	4
5 × n	5	10	15	20
term	9	14	19	24

1. **The difference is 5, so compare each term with the 5 times table.**
2. **Each term is 4 more than the number above it. The rule is 5n + 4.**
3. **Check the 4th term: 5 × 4 + 4 = 24. It matches.**
4. **The 100th term is 5 × 100 + 4 = 504.**
5. **Test it in code: set term to 9, then repeat 99 times: change term by 5. The 100th term is 99 jumps after the 1st term.**

Answer: The rule is 5n + 4. The 100th term is 504.

How to find the rule for the nth term

1. Find the difference between neighbouring terms.
2. Write that times table under n: 5, 10, 15, ...
3. Find how much to add or take away to reach each term.
4. Check the rule on 2 terms. Then use it for any term.

Your turn

Answer at the back of the book.

Try

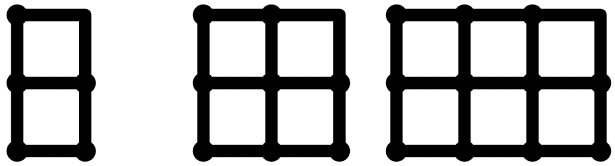
A sequence starts 3, 10, 17, 24, ...

The rule for the nth term is ☐ × n – ☐. The 100th term is ☐.

Start here

Do these first. Take your time.

1 Matchsticks make 2 rows of squares. Pattern 2 has 2 columns and pattern 3 has 3 columns.



Pattern 1 Pattern 2 Pattern 3

(a) Pattern 3 has ☐ sticks. Each new column adds ☐ sticks.

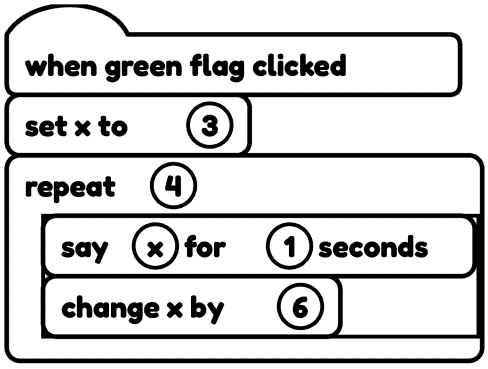
(b) The rule: sticks = ☐ × n + ☐

(c) Pattern 40 has ☐ sticks. (d) Pattern ☐ has 92 sticks.

2 Complete the table for each sequence.

Sequence	Difference	Rule for term n	Term 100
(a) 8, 14, 20, 26, …			
(b) 3, 8, 13, 18, …			

3 Trace the code. The sprite says x first, then changes it. Fill in the trace table.



Loop	x said	x after
1		
2		
3		
4		

(a) The last number it says is ☐.

(b) With repeat 50, the last number it says is ☐.

Start here

Keep going. Check your work.

4 **A sequence has the rule 5n + 2. Its first terms are 7, 12, 17, 22, ...**

(a) Is 157 a term? Yes No If yes, it is term number ☐.

(b) Is 205 a term? Yes No

(c) The 60th term is ☐.

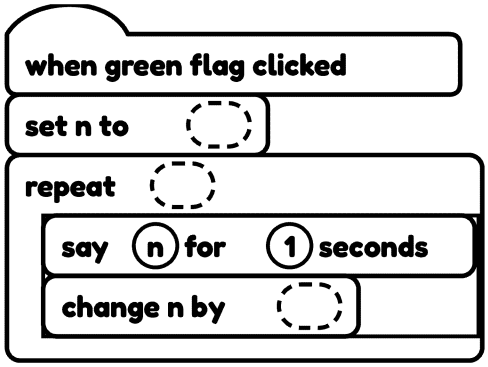
5 **Complete the code so that the sprite says the first 15 terms of 11, 18, 25, 32, ...**

(a) set n to ☐

(b) repeat ☐

(c) change n by ☐

(d) The last number it says is ☐.

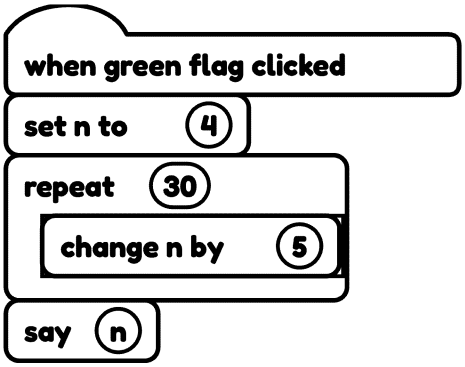


6 **Siti's code should say the 30th term of 4, 9, 14, 19, ...**

(a) Her code says ☐.

(b) The 30th term really is ☐.

(c) To fix it, change repeat 30 to repeat ☐.



7 **Circle the rule that gives each sequence. Test n = 1 and n = 2.**

(a) 4, 9, 14, 19, ... 4n + 1 5n – 1 5n + 4

(b) 10, 13, 16, 19, ... 3n + 10 10n + 3 3n + 7

(c) 3, 11, 19, 27, ... 8n + 3 8n – 5 3n + 8

Start here

Nearly there. You can do it.

8 **Ali has $15 on Day 1. Every day after that he adds $7. So on Day 2 he has $22.**

(a) On Day 4 he has $ ☐.

(b) Money on Day n = ☐ × n + ☐ dollars.

(c) On Day 30 he has $ ☐.

(d) He first has more than $200 on Day ☐.

9 **A program begins on the 1st term of a sequence. Each repeat moves it on to the next term.**

(a) To reach the 60th term, repeat ☐ times.

(b) After repeat 40, the sprite is at term number ☐.

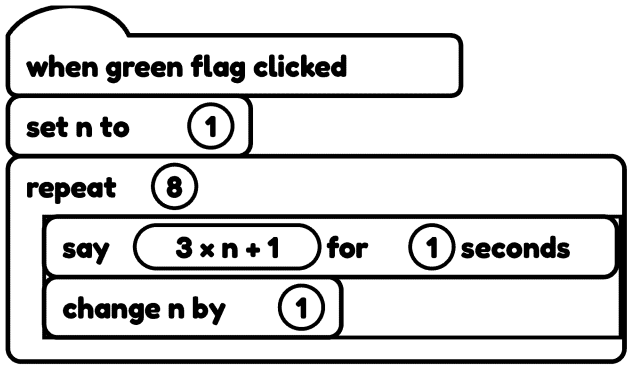
(c) To go from the 8th term to the 25th term, repeat ☐ times.

10 **This code uses n as a counter and says a rule. Predict what it says.**

(a) It starts: _____

(b) The last number it says is ☐.

(c) To say 5, 9, 13, ... use ☐ × n + ☐



11 **Aisha looks at 2, 8, 14, 20, ... The difference is 6, so she says the rule is 6n.**

(a) Is she right? Yes No

(b) The rule 6n gives ☐ for the 1st term, but the 1st term is ☐.

(c) The correct rule is ☐ × n − ☐.

Think harder

Hints are upside down. Try first!

1 **A sequence goes down by the same amount each time: 58, 53, 48, 43, …**

(a) The rule for the nth term is ☐ − 5n. **(b)** The 10th term is ☐.

(c) The last term above 0 is term number ☐.

Working: _____

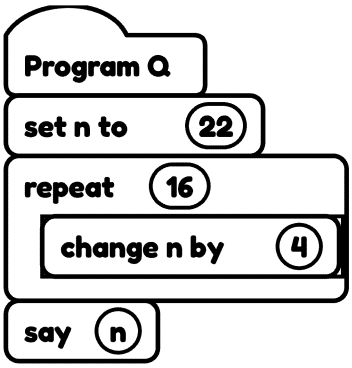
Hint. By how much do the terms change each step? Build the rule from it.

2 **Program P starts at the 1st term of 6, 10, 14, 18, … Program Q starts at the 5th term of the same sequence. Both change by 4 each repeat.**

(a) Q ends at term number ☐.

(b) To end at the same term, P must repeat ☐ times.

(c) Both programs say ☐.



Hint. Count how many terms Q moves on. P starts 4 terms earlier.

3 **A sequence goes up by the same amount each time. Its 4th term is 23 and its 9th term is 53.**

(a) From the 4th term to the 9th term is ☐ jumps.

(b) Each jump is ☐. **(c)** The 1st term is ☐.

(d) The rule is ☐ × n − ☐.

Working: _____

Hint. How many jumps are there between the 2 known terms?

★ Go further

A weekend page. Big challenges. Hints are upside down.

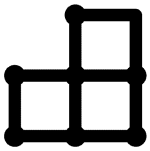
★ Beyond Level 6: Second differences

1

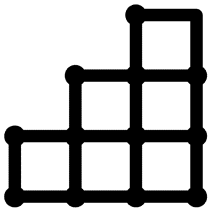
Matchsticks make staircases. Look at staircases 1, 2 and 3.



Staircase 1



Staircase 2



Staircase 3

(a) Count the sticks. Staircase 2 has ☐ and staircase 3 has ☐.

(b) The extra sticks go up by the same amount each time. Staircase 4 has ☐ sticks and staircase 5 has ☐.

(c) Staircase 10 has ☐ sticks.

Hint. Find how many extra sticks each staircase adds. Look for a pattern.

2

A sprite says these numbers in order: 3, 20, 7, 25, 11, 30, 15, 35, ...

(a) Places 1, 3, 5, ... go up by ☐. Places 2, 4, 6, ... go up by ☐.

(b) The number in place 15 is ☐.

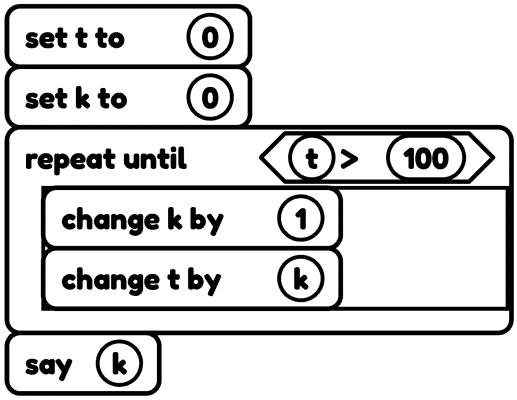
(c) The number in place 20 is ☐.

Hint. Split it into 2 sequences. Which one is place 15 in?

★ Beyond Level 6: Triangle numbers

3

What does this code say? Write t after each repeat.



(a) The code says ☐.

(b) When it stops, t is ☐.

Hint. Write t after every repeat: 1, 3, 6, ... Where does it pass 100?

✔ Check yourself

No hints. Try each one on your own.

1 Find the rule for 9, 16, 23, 30, … The rule is ☐ × n + ☐. The 100th term is ☐.

Working: _____

2 Is 125 a term of 7, 12, 17, 22, …? Yes No

Working: _____

3 The code is: set n to 8, then repeat 20 times: change n by 6. At the end n = ☐, which is term number ☐ of 8, 14, 20, …

Working: _____

4 Mei starts at n = 5. She repeats 4 times: change n by 3, then say n. What is the first number it says? ☐

Working: _____

My score: ☐ **out of 4. Check the answers at the back.**

What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a helper. Then try Check yourself again.

Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

Learn it: Primes and factors

My goal. I can find all the factors of a number, test whether it is prime and write it as a product of primes. I can count factors with a loop.

A factor divides a number exactly. Factors come in pairs, and a prime has exactly 2 factors: 1 and itself. In code, n mod d is the remainder, and a remainder of 0 means d is a factor.

Worked example

Is 97 prime? Show your test.

d	2	3	5	7
97 mod d	1	1	2	6

1. **A prime has only 2 factors. We look for any factor besides 1 and 97.**
2. **Test the primes 2, 3, 5 and 7. Stop at 7, because 11 × 11 = 121 is already more than 97.**
3. **Find the remainder 97 mod d for each. None of them is 0.**
4. **A program that counts the factors of 97 ends with count = 2.**

Answer: 97 is prime: it has no factor except 1 and 97.

How to test whether n is prime

1. Try the primes 2, 3, 5, 7, … in order.
2. Find n mod d. A remainder of 0 means d is a factor.
3. Stop when d × d is more than n.
4. No remainder was 0? Then n is prime.
5. Quick tests: an even last digit means 2 is a factor. Digits adding to a multiple of 3 mean 3 is a factor. A last digit of 0 or 5 means 5 is a factor.
6. Factor tree: split a number into 2 factors again and again. The ends that cannot split are primes: 60 = 6 × 10 = 2 × 3 × 2 × 5.

Your turn

Answer at the back of the book.

Try **Is 143 prime?** Yes No **If not, 143 =** ☐ × ☐ .

Start here

Do these first. Take your time.

1 **Factors come in pairs. Use pairs to find all the factors.**

(a) The factors of 42 are _____

(b) 100 has ☐ factors.

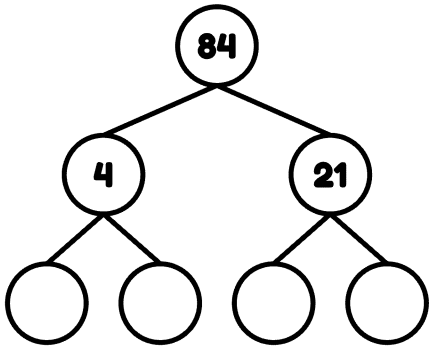
(c) The factor of 100 that pairs with itself is ☐.

2 **Circle every prime number in the table.**

51	53	57	59	63
67	69	73	77	79

3

Finish the factor tree of 84. Split each number until only primes are left.



(a) 84 = ☐ × ☐ × ☐ × ☐

(b) Write 90 as a product of primes:

_____

4 **Write 26 as a sum of 2 primes. The 2 primes may be the same. Find every way.**

2	3	5	7	11	13	17	19	23	29

_____

Start here

Keep going. Check your work.

5 **This code counts the factors of n. Fill in the trace table.**

```
set n to 20
set d to 1
set count to 0
repeat 6
    if n mod d = 0 then
        change count by 1
    change d by 1
say count
```

d	n mod d	count
1		
2		
3		
4		
5		
6		

(a) The code says ☐.

(b) To count all the factors of 20, change repeat 6 to repeat ☐.

6 **Who am I? Each riddle has only one answer.**

(a) I am a prime number between 40 and 60. My digits add to 8. I am ☐.

(b) I am a prime number between 60 and 80. If you add 2 to me, you get another prime. I am ☐.

7 **Most numbers have an even number of factors. Some do not.**

(a) 81 has ☐ factors.

(b) The factor of 81 that pairs with itself is ☐.

(c) Which has an odd number of factors? 50 64 72

Start here

Nearly there. You can do it.

8 **Use the divisibility tests. Circle Yes or No.**

(a) Is 372 a multiple of 4? Yes No

(b) Is 2457 a multiple of 3? Yes No

(c) Is 6835 a multiple of 9? Yes No

9 **Test whether 113 is prime. You only need to test 2, 3, 5 and 7.**

(a) 113 mod 2 = ☐, mod 3 = ☐, mod 5 = ☐, mod 7 = ☐

(b) Is 113 prime? Yes No

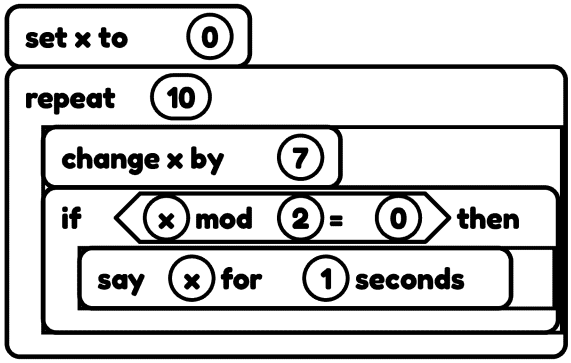
(c) Is 119 prime? Yes No If not, 119 = ☐ × ☐.

10 **Predict what this code says.**

(a) The first number it says is ☐.

(b) It says ☐ numbers.

(c) The last number it says is ☐.



11 **Ali has 36 cards. He puts them in equal rows, with at least 2 rows and at least 2 cards in each row.**

(a) The number of rows could be

_____

(b) There are ☐ different ways.

Think harder

Hints are upside down. Try first!

1 **Find the smallest number with exactly this many factors.**

(a) 5 factors: ☐ **(b)** 9 factors: ☐ **(c)** 10 factors: ☐

Working: _____

_____

Hint. Check numbers in order. Count their factors until you reach 5.

2 **A number is 2 × 3 × p, where p is a prime greater than 5. The number is less than 100.**

(a) The possible numbers are _____

(b) Each of them has ☐ factors.

(c) Which of them is a multiple of 11? 42 66 78

Working: _____

_____

Hint. Try p = 7, then the next primes. Stop at 100.

3 **Find all the primes between 100 and 120.**

(a) The primes are _____

(b) Pairs of primes that differ by 2: ☐ pairs.

(c) Two odd numbers between 100 and 120 are multiples of 7. They are ☐ and ☐.

Working: _____

_____

Hint. Cross out even numbers first. Then test 3, 5 and 7.

★ Go further

A weekend page. Big challenges. Hints are upside down.

★ Beyond Level 6: Counting factors

1 Break a number into primes, then count its factors. Each factor uses some of the primes.

(a) List the factors of 72, then count them: ☐

(b) 180 = 2 × 2 × 3 × 3 × 5 has ☐ factors.

(c) The smallest number with exactly 12 factors is ☐.

Factors of 72: _____

Hint. Break 72 into primes first. Then count the choices for each prime.

2 Use the sieve to find the primes up to 60. Start by crossing out 1. Then do this for 2, 3, 5 and 7: ring the number and cross out its other multiples.

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50
51	52	53	54	55	56	57	58	59	60

(a) There are ☐ primes up to 60. **(b)** The 10th prime is ☐.

Hint. Why can you stop at 7? Think of 8 × 8.

3 This code looks for the smallest factor of n that is greater than 1. Run it.

(a) With n = 221 it says ☐.

(b) With n = 221 the repeat runs ☐ times.

(c) With n = 97 it says ☐.

```
set d to (2)
repeat until (n) mod (d) = (0)
    change d by (1)
say (d)
```

Hint. Write d each time. It stops when n mod d is 0.

✔ Check yourself

No hints. Try each one on your own.

1 **List all the factors of 40.** _____

Working: _____

2 **Is 147 prime?** **Yes** **No**

Working: _____

3 **Write 140 as a product of primes.**

_____

Working: _____

4 **47 mod 6 =** ☐ **and 100 mod 8 =** ☐ **.**

Working: _____

My score: ☐ **out of 4. Check the answers at the back.**

What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a helper. Then try Check yourself again.

Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

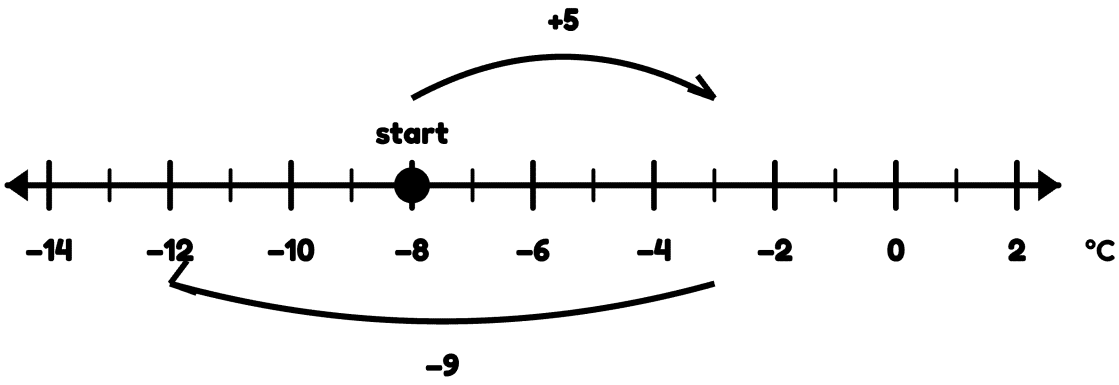
Learn it: Negative numbers

My goal. I can order negative numbers and add and subtract by jumping along a number line. I can follow a variable that goes below 0 in code.

Numbers below 0 have a minus sign. On a number line, greater numbers are to the right. To add, jump right. To subtract, jump left. When a jump crosses 0, split it at 0.

Worked example

A freezer thermometer reads –8 °C. The door is opened and it rises 5 degrees. Then the door is shut and it falls 9 degrees. What does it read now?



1. **Start at –8. Rising moves right: 5 steps to –3.**
2. **Falling moves left. From –3, 3 steps reach 0, then 6 more steps.**
3. **The thermometer reads –12 °C.**
4. **Check by adding: –8 + 5 – 9 = –12.**

Answer: It reads –12 °C.

How to add and subtract with negative numbers

1. Mark the start on the number line.
2. Adding moves right. Subtracting moves left.
3. If the jump crosses 0, split it at 0.
4. Check by jumping back to the start.

Your turn

Answer at the back of the book.

Try **It is –3 °C. It falls 8 degrees. What is the temperature now?** ☐ °C

Start here

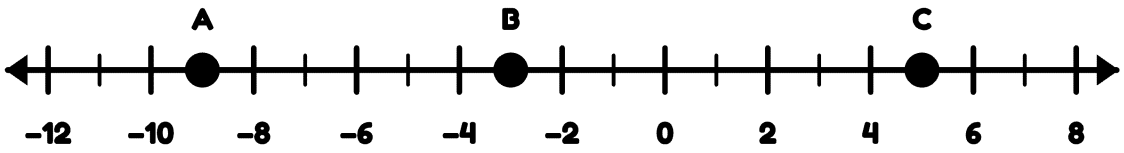
Do these first. Take your time.

1 **Put the numbers in order. Begin with the smallest.**

(a) 2, −6, −11, 0, −3, 8 gives _____

(b) Circle > or <: −4 > < −10 0 > < −2 −15 > < 1

2 **Read the number line. Each mark is 1 step.**



(a) A = ☐ B = ☐ C = ☐

(b) From A to B is ☐ steps. From B to C is ☐ steps.

3 **The thermometer shows 3 readings on one day.**

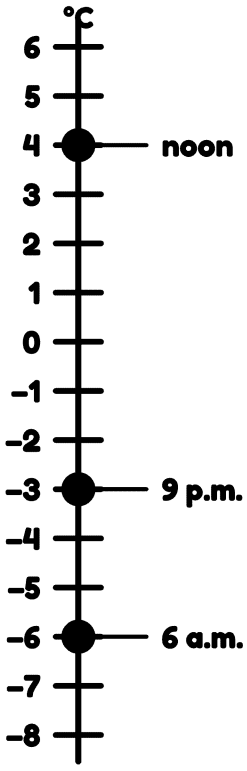
(a) At 6 a.m.: ☐ °C

(b) At noon: ☐ °C

(c) At 9 p.m.: ☐ °C

(d) From 6 a.m. to noon it rose ☐ degrees.

(e) From noon to 9 p.m. it fell ☐ degrees.



Start here

Keep going. Check your work.

4 **Find the value of each one. Use a number line to help.**

(a) $-6 + 10 =$ ☐ (b) $3 - 9 =$ ☐ (c) $-7 - 5 =$ ☐

(d) $-4 + 15 =$ ☐

5 **Find the missing number.**

(a) $-5 +$ ☐ $= 6$ (b) $4 -$ ☐ $= -3$ (c) ☐ $+ 8 = 2$

(d) ☐ $- 5 = -9$

6 **How far is it from the first number to the second? Count the steps, splitting at 0.**

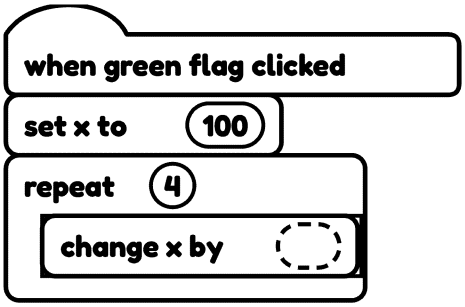
(a) -9 to 4: ☐ steps (b) -6 to -1: ☐ steps

(c) -14 °C to 31 °C: ☐ degrees

7 **A sprite starts at x = 100. Fill in the blocks.**

(a) It ends at x = 0. change x by ☐

(b) With repeat 8 and change x by (−20), it ends at x = ☐.



8 **Think about how far each number is from 0.**

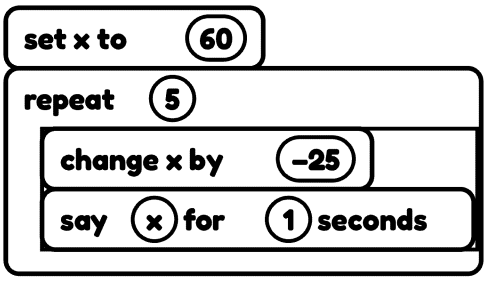
(a) Circle the number closest to 0: -9 5 -3 7 -4

(b) Which is further from 0? -8 6

Start here

Nearly there. You can do it.

9 **Trace the code. The sprite changes x, then says it.**



Loop	x said
1	
2	
3	
4	
5	

(a) The 3rd number it says is ☐.

(b) The last number it says is ☐.

(c) It says ☐ numbers that are below 0.

10 **A balance of −15 dollars means Ali owes 15 dollars. Ali has 25 dollars.**

(a) He buys a game for 40 dollars. His balance is ☐ dollars.

(b) Then he is given 30 dollars. His balance is ☐ dollars.

(c) To go from −15 dollars up to 20 dollars, he needs ☐ dollars.

11 **Heights are measured from sea level, 0 m. A cave is at −35 m, a cliff top is at 48 m and a boat is at −6 m.**

(a) The cliff top is ☐ m above the cave.

(b) The boat is ☐ m above the cave.

(c) A bird is 12 m above the cliff top. It is at ☐ m.

Think harder

Hints are upside down. Try first!

★ Beyond Level 6: Subtracting a negative number

1 **To find 6 – (–4), count the steps from –4 up to 6.**

(a) 6 – (–4) = ☐ **(b)** –2 – (–5) = ☐

(c) So taking away –4 gives the same result as adding ☐.

Number line: _____

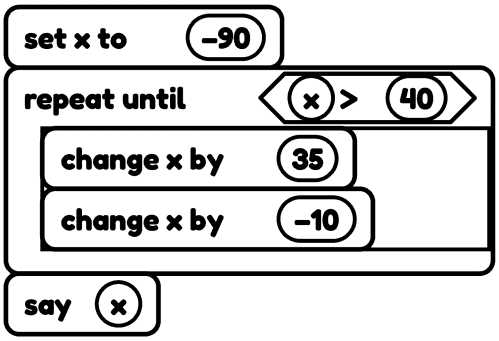
_____

Hint. Picture both numbers on a number line.

2 **A sprite starts at x = –90. The loop stops as soon as x is greater than 40 at the start of a repeat.**

(a) The loop repeats ☐ times.

(b) The sprite says ☐.



Hint. Write x after each repeat. Is it more than 40 yet?

3 **On Monday at noon it was 3 °C. On Tuesday it was 11 degrees colder than Monday. On Wednesday it was 6 degrees warmer than Tuesday. On Thursday it was –4 °C.**

(a) Tuesday: ☐ °C **(b)** Wednesday: ☐ °C

(c) From Monday to Thursday it fell ☐ degrees.

Number line: _____

_____

Hint. Draw a number line. Mark Monday first, then each change.

★ Go further

A weekend page. Big challenges. Hints are upside down.

1 **Two divers: A is at −12 m and goes down 4 m every minute. B is at −40 m and comes up 6 m every minute.**

(a) After 3 minutes, A is at ☐ m and B is at ☐ m.

(b) B is first above A after minute ☐.

(c) After 5 minutes they are ☐ m apart.

Working: _____

_____

Hint. Make a table: minute, depth of A, depth of B.

2 **This code uses a and b. Run it for each pair of numbers.**

(a) a = −8 and b = 5. It says ☐.

(b) a = −3 and b = −9. It says ☐.

(c) a = −6 and b = −6. It says ☐.

```
if (a > b) then
    say (a − b)
else
    say (b − a)
```

Hint. Is a greater than b? Choose the right branch first.

★ **Beyond Level 6: A calendar with no year 0**

3 **There is no year 0. The year after 1 BC is AD 1. Count the years one by one.**

(a) From 5 BC to 1 BC is ☐ years.

(b) From 3 BC to AD 4 is ☐ years.

(c) A person born in 10 BC dies in AD 12. They are ☐ years old.

Working: _____

_____

Hint. Which year comes straight after 1 BC? Count carefully.

✔ Check yourself

No hints. Try each one on your own.

1 **−9 + 14 =** ☐ **6 − 13 =** ☐

Working: _____

2 **Put in order, smallest first: 4, −7, −2, 0**

_____

Working: _____

3 **It is −6 °C. It rises 11 degrees, then falls 8 degrees. It is now** ☐ **°C.**

Working: _____

4 **A sprite starts at x = 20. It runs repeat 3 [change x by −15]. Then x =** ☐ **.**

Working: _____

My score: ☐ **out of 4. Check the answers at the back.**

What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a helper. Then try Check yourself again.

Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

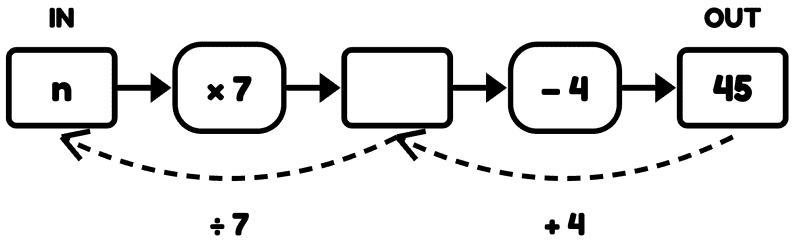
Learn it: Machines and equations

My goal. I can find the rule of a number machine from a table and write it with a letter. I can run a machine backwards to solve an equation.

A number machine follows a rule: IN goes in and OUT comes out. We write the rule with a letter, such as OUT = 4n + 3. To find the IN for a given OUT, run the machine backwards and undo the last step first.

Worked example

A machine does × 7, then − 4. OUT is 45. What was IN?



1. **The rule is OUT = 7n − 4, so 7n − 4 = 45.**
2. **Undo the last step first. Undo take away 4 by adding 4: 45 + 4 = 49.**
3. **Then undo × 7 by dividing by 7: 49 ÷ 7 = 7.**
4. **Check forwards: 7 × 7 = 49, then 49 − 4 = 45.**

Answer: IN was 7.

How to run a machine backwards

1. Write the steps in order, from IN to OUT.
2. Start at OUT. Undo the last step first.
3. Use the opposite step: + undoes −, and × undoes ÷.
4. Check by running the machine forwards.

Reading the code blocks

- repeat 5: do the inside 5 times
- repeat until <test>: stop when true
- if <test> then / else: choose a part
- n mod d: remainder of n ÷ d
- pick random 1 to 6: like a die
- item i of list: the i-th item
- answer: what the user typed in
- ↻ turn: turn clockwise

Your turn

Answer at the back of the book.

Try **A machine does × 4, then + 9. OUT is 41. What was IN?** ☐

Start here

Do these first. Take your time.

1 **Guess my rule. The rule is: multiply n by a number, then add a number.**

IN	1	3	5	7
OUT	5	11	17	23

(a) OUT = ☐ × n + ☐ (b) IN 10 gives OUT ☐.

(c) OUT 50 comes from IN ☐.

Working: _____

2 **Guess my rule. This time the machine takes a number away.**

IN	2	4	6	8
OUT	1	7	13	19

(a) OUT = ☐ × n – ☐ (b) IN 20 gives OUT ☐.

(c) OUT 100 comes from IN ☐.

Working: _____

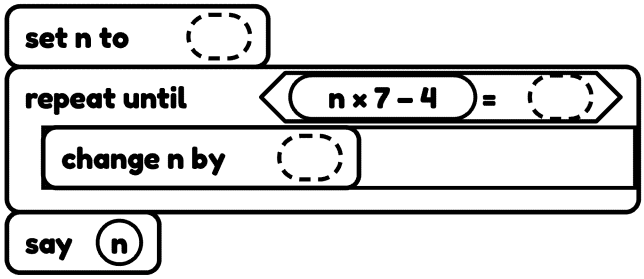
3 **This loop hunts for the n that makes 7n – 4 equal 73. It tries 0, 1, 2, … in turn. Complete the blocks.**

(a) set n to ☐

(b) repeat until it is ☐

(c) change n by ☐

(d) It says ☐.



Start here

Keep going. Check your work.

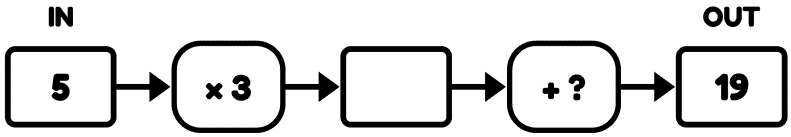
4 **Machine P does × 2, then + 6. Machine Q does + 6, then × 2. Use the same IN for both.**

(a) IN 7: P gives ☐ and Q gives ☐.

(b) IN 10: P gives ☐ and Q gives ☐.

(c) Do P and Q give the same OUT? Yes No

5 **A machine does 2 steps. Look at the first IN and OUT.**



(a) The middle box is ☐. (b) The last step is add ☐.

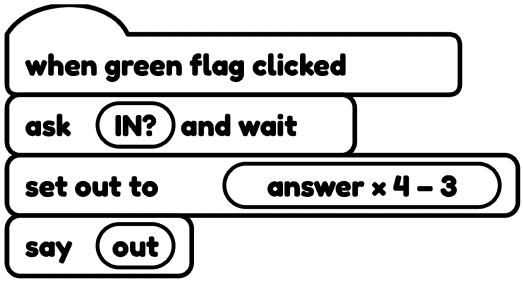
(c) IN 9 gives OUT ☐.

6 **The block "answer" holds the number you type. In code, × means times. Predict what the sprite says.**

(a) You type 9. It says ☐.

(b) You type 25. It says ☐.

(c) It says 61. You typed ☐.



7 **Siti has n marbles. Kumar has 5 marbles more. Jenny has 3 times as many marbles as Siti.**

(a) When n = 8, Kumar has ☐ and Jenny has ☐.

(b) When n = 8, all 3 together have ☐ marbles.

(c) All 3 together have ☐ × n + ☐ marbles.

Start here

Nearly there. You can do it.

8 **Find n in each equation by undoing the steps. Then check by going forwards.**

(a) $5n + 8 = 63$ n = ☐ **(b)** $9n - 7 = 74$ n = ☐

(c) $(n + 4) \times 6 = 66$ n = ☐ **(d)** $n \div 5 + 3 = 11$ n = ☐

Check: _____

9 **A phone plan costs 12 dollars a month, plus 3 dollars for each extra GB. n is the number of extra GB.**

(a) 5 extra GB cost ☐ dollars.

(b) The cost is ☐ × n + ☐ dollars.

(c) With 60 dollars you can buy ☐ extra GB.

10 **Pick a number. Add 4. Multiply by 3. Subtract 9. Divide by 3. Subtract the number you picked.**

(a) Start with 7. You get ☐.

(b) Start with 12. You get ☐.

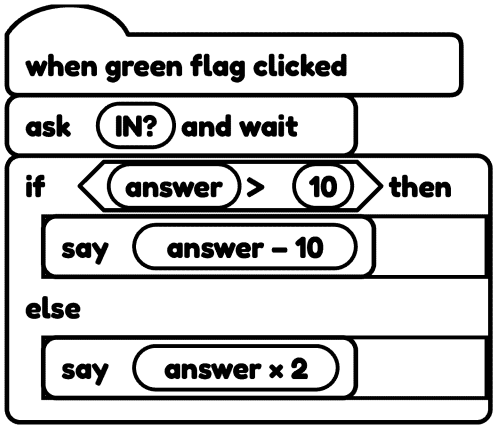
(c) Start with n. The steps give n + 4, 3n + 12, 3n + 3, n + 1, then ☐.

11 **Predict what the sprite says.**

(a) You type 6. It says ☐.

(b) You type 15. It says ☐.

(c) Two different numbers make it say 8.
They are ☐ and ☐.



Think harder

Hints are upside down. Try first!

1 **Machine X does × 2, then + 3. Its OUT goes straight into Machine Y, which does × 3, then − 4.**

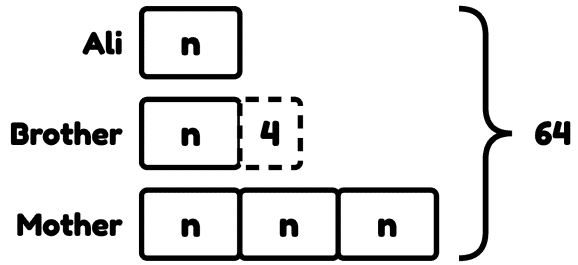
(a) IN 5 gives OUT ☐. **(b)** OUT 77 comes from IN ☐.

(c) Joined, the rule is OUT = ☐ × n + ☐.

Working: _____

Hint. Which end of the joined machines do you already know?

2 **Ali is n years old. His brother is 4 years older. Their mother is 3 times as old as Ali. Together the 3 of them are 64 years old.**



(a) Together: ☐ × n + ☐ = 64 **(b)** Ali is ☐ years old.

(c) Their mother is ☐ years old.

Hint. Add up the n boxes first. Then add the extra 4.

3 **A machine multiplies IN by a whole number, then adds or subtracts a whole number. IN 3 gives OUT 13 and IN 8 gives OUT 38.**

(a) When IN goes up by 5, OUT goes up by ☐.

(b) The rule is OUT = ☐ × n − ☐.

(c) OUT 98 comes from IN ☐.

Working: _____

Hint. How much does OUT go up for each 1 that IN goes up?

★ Go further

A weekend page. Big challenges. Hints are upside down.

★ Beyond Level 6: n on both sides

1 Machine A does × 4, then + 1. Machine B does × 2, then + 13.

(a) IN 2: A gives ☐ and B gives ☐.

(b) A and B give the same OUT when IN is ☐. That OUT is ☐.

IN	A	B

Hint. Make a table of IN, A and B. Look at how the gap changes.

2 Call the smallest number n.

(a) 4 whole numbers that follow each other make a total of 130. The smallest is ☐.

(b) 5 odd numbers that follow each other make a total of 125. The smallest is ☐.

(c) Can 4 whole numbers that follow each other make a total of 100?

Yes No

Working: _____

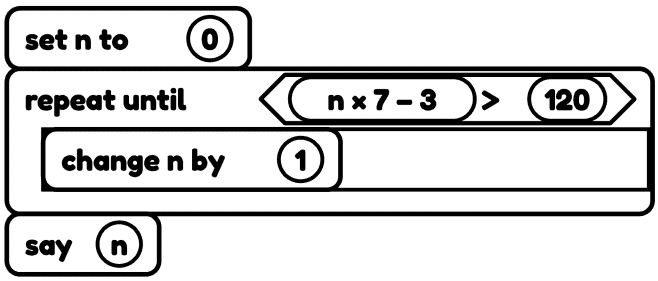
Hint. Write each number with n. Add them up.

3 A search program tries n = 0, 1, 2, … It stops when 7n – 3 is greater than 120.

(a) It says ☐.

(b) When it stops, 7n – 3 is ☐.

(c) If 120 changes to 123, it says ☐.



Hint. Is 7n – 3 greater than 120? Careful with equal.

✔ Check yourself

No hints. Try each one on your own.

1 **IN 1, 2, 3, 4 give OUT 6, 11, 16, 21. The rule is OUT = ☐ × n + ☐.**

Working: _____

2 **A machine does × 9, then − 5. OUT is 49. IN = ☐**

Working: _____

3 **4n + 9 = 57. n = ☐**

Working: _____

4 **Ali has n sweets. Mei has 4 more. Together they have 30. n = ☐**

Working: _____

My score: ☐ out of 4. Check the answers at the back.

What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a helper. Then try Check yourself again.

Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

Learn it: Counting

My goal. I can count every possible case without missing one or counting one twice. I can use multiplication for choices made one after another.

To count every possible case, list them in order and do not miss one. Choices made one after another multiply. When the order does not matter, each pair is counted twice, so halve the total.

Worked example

A cafe sells 3 sandwiches and 4 drinks. A meal is 1 sandwich and 1 drink. How many different meals are there?

	Tea	Juice	Milk	Water
Egg	1	2	3	4
Ham	5	6	7	8
Tuna	9	10	11	12

1. Fix one sandwich. It goes with each of the 4 drinks, so there are 4 meals.
2. There are 3 sandwiches, so there are 3 groups of 4 meals.
3. 3 × 4 = 12 meals. Each cell of the table is one meal.
4. Add a choice of 2 fruits. Every meal splits in 2 ways: 12 × 2 = 24.

Answer: 12 meals (24 meals if there is also a choice of 2 fruits).

How to count every possible way

1. List the choices in order. Fix the first choice, then run through the next.
2. Choices made one after another multiply: 3 × 4.
3. If the order does not matter, each pair is counted twice. So halve.
4. Check a small case by listing it all.

Your turn

Answer at the back of the book.

Try A pizza place has 4 crusts and 6 toppings. A pizza has 1 crust and 1 topping.

How many different pizzas are there? ☐

Start here

Do these first. Take your time.

1 **A bike shop builds a bike from 1 frame, 1 seat and 1 bell. There are 5 frames, 3 seats and 2 bells.**

(a) There are ☐ different bikes.

(b) The shop adds 1 more frame. Now there are ☐ bikes.

(c) Red is one of the 5 frames. ☐ bikes have a red frame.

2 **Flip a coin and roll a dice. A result looks like H3 (heads and 3). Use the table to list the results.**

	1	2	3	4	5	6
H						
T						

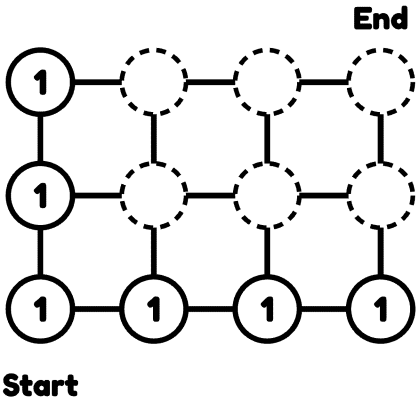
(a) There are ☐ different results.

(b) Tails and a number greater than 4: ☐ results.

(c) Heads and an even number: ☐ results.

3 **A walker goes only right or up from Start to End. Each number shows how many routes reach that corner. Fill in the empty circles.**

(a) There are ☐ routes from Start to End.



Start here

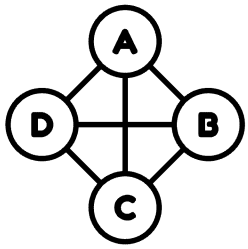
Keep going. Check your work.

4 **In a chess club, any 2 players play exactly 1 game against each other.**

(a) With 8 players there are ☐ games.

(b) With 11 players there are ☐ games.

(c) A club plays 66 games. It has ☐ players.



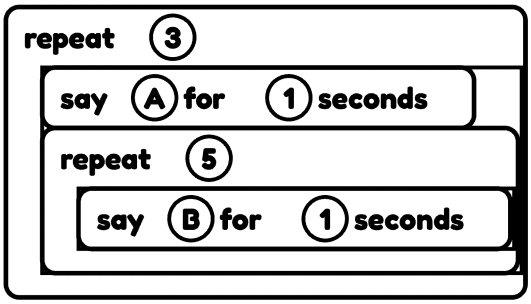
4 players: 6 games

5 **The inner loop does all its repeats each time the outer loop goes round. Predict what the sprite says.**

(a) It says A ☐ times.

(b) It says B ☐ times.

(c) In all it says ☐ things.



6 **A password has 3 letters. Each letter is A, B, C or D.**

(a) Letters may repeat: ☐ passwords.

(b) No letter may repeat: ☐ passwords.

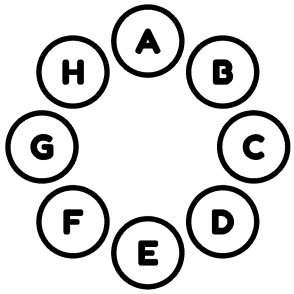
(c) Letters may repeat, and the first letter is A: ☐ passwords.

7 **An octagon has 8 corners. Join every pair of corners with a straight line.**

(a) There are ☐ lines in all.

(b) Diagonals (lines that are not sides): ☐

(c) A shape with 10 corners has ☐ diagonals.



Start here

Nearly there. You can do it.

8 **There are 5 runners in a race. Gold, silver and bronze go to 3 different runners.**

(a) The medals can be given in ☐ different ways.

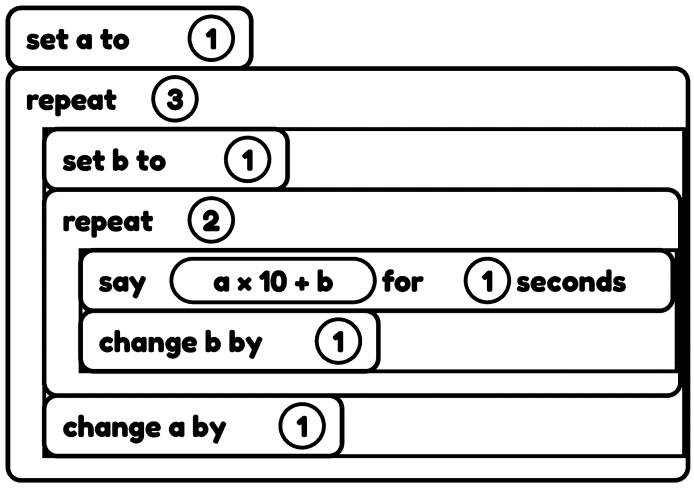
(b) Ali is one of the runners and must win gold. Then there are ☐ ways.

9 **This code lists pairs. The outer repeat changes a and the inner repeat changes b. Predict what it says.**

(a) The 3rd number it says is ☐.

(b) It says ☐ numbers.

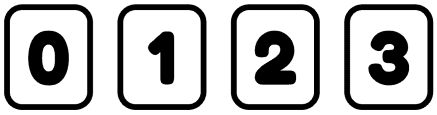
(c) With repeat 4 as the inner repeat, it says ☐ numbers.



10 **Make 3-digit numbers from the cards, using no card twice. A number cannot start with 0.**

(a) There are ☐ different numbers.

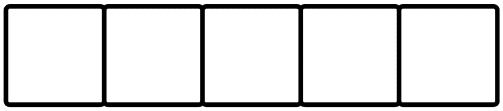
(b) ☐ of them are odd.



11 **A rectangle in a row of squares can be 1, 2, 3, … squares long. Count every rectangle.**

(a) This row of 5 squares has ☐ rectangles.

(b) A row of 7 squares has ☐ rectangles.

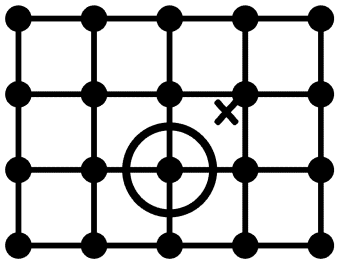


Think harder

Hints are upside down. Try first!

1

A walker goes only right or up from Start to End. X is a point on the way.



(a) Routes from Start to X: ☐. Routes from X to End: ☐.

(b) Routes that go through X: ☐.

Hint. A route through X has 2 parts. Count each part.

2

You have 4 digit cards: 1, 2, 3, 4. Put all 4 cards in a row to make a 4-digit number.

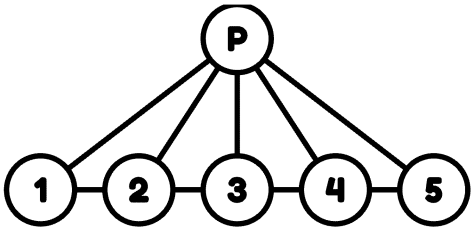
(a) There are ☐ numbers in all. (b) ☐ of them are even.

(c) ☐ of them are multiples of 4.

Hint. Which two-digit endings are in the 4 times table?

3

P is not on the line. A triangle uses P and 2 of the points on the line.



(a) With 4 points on the line there are ☐ triangles.

(b) With 5 points there are ☐ triangles.

(c) With 8 points there are ☐ triangles.

Hint. Each triangle is a pair of points on the line. Count pairs.

★ Go further

A weekend page. Big challenges. Hints are upside down.

★ Beyond Level 6: Count the opposite

1 **A code has 3 digits. Each digit is 1, 2, 3, 4 or 5, and digits may repeat.**

(a) There are ☐ codes in all.

(b) ☐ codes have no 5.

(c) ☐ codes have at least one 5.

Hint. Count the codes with no 5. Take them from the total.

2 **This code checks every pair of numbers a and b. Predict what it says.**

```
set count to 0
set a to 1
repeat 4
    set b to 1
    repeat 4
        set p to a × b
        if p mod 2 = 0 then
            change count by 1
        change b by 1
    change a by 1
say count
```

(a) It checks ☐ pairs.

(b) It says ☐.

(c) Both repeats change to 6. It says ☐.

Hint. When is a × b odd? Count those pairs first.

★ Beyond Level 6: A counting chain

3 **A child climbs a staircase. Each move goes up 1 step or 2 steps. Count the ways to reach the top.**

(a) 3 steps: ☐ ways. (b) 4 steps: ☐ ways.

(c) 6 steps: ☐ ways. (d) 8 steps: ☐ ways.

Hint. Look at the last move. It was 1 step or 2 steps.

✔ Check yourself

No hints. Try each one on your own.

1 **A cafe has 5 soups and 3 breads. A meal is 1 soup and 1 bread. There are ☐ different meals.**

Working: _____

2 **In a league of 5 teams, each team meets each of the others at home and away. There are ☐ matches.**

Working: _____

3 **Cards 2, 5 and 8 make 3-digit numbers when each card is used once. How many numbers? ☐**

Working: _____

4 **A code is 2 letters from A to E, then 1 digit from 1 to 4. Letters and digits may repeat. There are ☐ codes.**

Working: _____

My score: ☐ out of 4. Check the answers at the back.

What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a helper. Then try Check yourself again.

Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

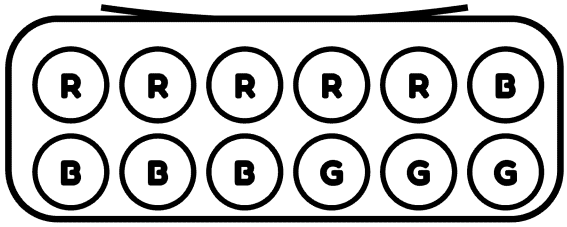
Learn it: Probability

My goal. I can write a probability as a fraction, decide whether a game is fair, and say what to expect from many tries.

A probability is the number of ways you want, out of all the equally likely outcomes. A game is fair when every player has the same probability of winning. In many tries we expect about probability × tries, but a real run varies.

Worked example

A bag has 12 counters: 5 red (R), 4 blue (B) and 3 green (G). One is picked without looking. What is the probability of blue? About how many blues in 600 picks, if the counter goes back each time?



1. **All 12 counters are equally likely to be picked.**
2. **4 of the 12 are blue, so P(blue) = 4/12 = 1/3.**
3. **In 600 picks we expect 1/3 × 600 = 200 blues.**
4. **A real run gives a number near 200, not exactly 200. Each pick is random.**

Answer: P(blue) = 1/3. About 200 blues in 600 picks.

How to find a probability

1. Count all the equally likely outcomes.
2. Count the outcomes you want.
3. Write wanted/all as a fraction and simplify it.
4. Expected number = probability × number of tries. A real run is close, not exact.

Your turn

Answer at the back of the book.

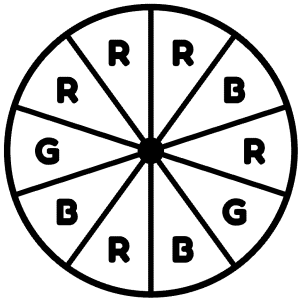
Try **A bag has 5 white and 15 black counters. One is picked without looking.**

P(white) = ☐ / ☐ in simplest form.

Start here

Do these first. Take your time.

1 **The spinner has 10 equal parts. R is red, B is blue and G is green. Give fractions in simplest form.**



(a) P(green) = ☐ / ☐

(b) P(not red) = ☐ / ☐

(c) In 500 spins, about ☐ are blue.

2 **A coin is flipped and a spinner with 3 equal parts A, B and C is spun. A result looks like HB. Use the table to list the results. Give fractions in simplest form.**

	A	B	C
H			
T			

(a) There are ☐ different results.

(b) P(heads and A) = 1/☐.

(c) P(tails and not A) = ☐ / ☐

3 **This code rolls a dice 100 times and counts the sixes. Predict what happens.**

(a) The chance of a 6 on one roll is ☐ / ☐.

(b) To the nearest whole number, the code says about ☐.

(c) 5 runs give 14, 19, 17, 21, 15. Why?

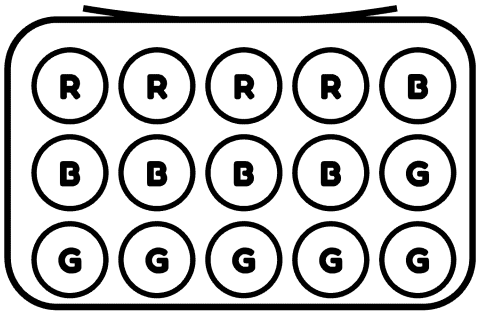
random broken
rolls code

```
set hits to 0
repeat 100
    if pick random 1 to 6 = 6 then
        change hits by 1
say hits
```

Start here

Keep going. Check your work.

4 **One counter is picked from the bag. R is red, B is blue and G is green. Give fractions in simplest form.**

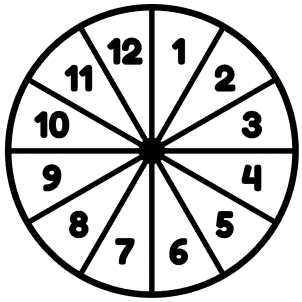


(a) P(blue) = ☐ / ☐

(b) P(red or green) = ☐ / ☐

(c) 5 more red counters are put in. Now P(red) = ☐ / ☐.

5 **Ali and Ben spin this spinner with the numbers 1 to 12. Ali wins on a multiple of 3. Ben wins on any other number.**



(a) P(Ali wins) = ☐ /12

(b) P(Ben wins) = ☐ /12

(c) Is the game fair? Yes No

6 **A dice is rolled 300 times and a coin is flipped 90 times. Give the number you expect.**

(a) About ☐ fives. (b) About ☐ numbers greater than 4.

(c) About ☐ heads in 90 flips.

(d) A run gives 54 fives. Is that surprising? Yes No

7 **Bag A has 3 red and 5 blue counters. Bag B has 4 red and 7 blue counters. You want to pick a red counter.**

(a) In Bag A, P(red) = ☐ / ☐.

(b) In Bag B, P(red) = ☐ / ☐.

(c) Which bag gives the better chance of red? Bag A Bag B

Start here

Nearly there. You can do it.

8 **Two dice, red and blue, are rolled and added. The table shows all 36 equally likely totals. Give fractions in simplest form.**

(a) ☐ of the 36 totals are 4.

(b) ☐ of the 36 totals are less than 5.

(c) P(total is 4) = ☐ / ☐

+	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	8
3	4	5	6	7	8	9
4	5	6	7	8	9	10
5	6	7	8	9	10	11
6	7	8	9	10	11	12

9 **Cards are numbered 1 to 30. One card is drawn without looking. Give fractions in simplest form.**

(a) P(multiple of 7) = ☐ / ☐

(b) P(prime) = ☐ / ☐

(c) P(square number) = ☐ / ☐

10 **Ravi flips a fair coin 200 times and gets 112 heads.**

(a) About ☐ heads are expected.

(b) The difference from the expected number is ☐.

(c) Could the coin still be fair? Yes No

11 **Two dice are rolled and added. The totals 2 to 12 are not equally likely.**

(a) The most likely total is ☐.

(b) P(total is 6 or 8) = ☐ /36

(c) P(total is 2 or 12) = ☐ /36

Think harder

Hints are upside down. Try first!

1 **Two dice are rolled and added. In Game 1, Ali wins on an even total and Ben wins on an odd total.**

(a) Game 1: Ali wins on ☐ of the 36 totals.

(b) Is Game 1 fair? Yes No

(c) Game 2: Ali wins on a total of 7. Ben wins on every other total and scores 1 point for a win. For a fair game Ali must score ☐ points for a win.

Working: _____

Hint. Fair means the winning points are equal over many games.

2 **A bag has 3 red counters (R1, R2, R3) and 2 blue counters (B1, B2). Two counters are taken out one after another. The first is not put back. Give fractions in simplest form.**

(a) There are ☐ equally likely ordered pairs.

(b) ☐ of them are both red. **(c)** P(both red) = ☐ / ☐

Working: _____

Hint. List the pairs: R1R2, R1R3, R1B1, … Is R2R1 different?

3 **A program rolls a dice 600 times and counts the 3s. Five runs give 93, 108, 97, 114 and 102.**

(a) The expected number of 3s is ☐.

(b) The mean of the 5 runs is ☐.

(c) Does the dice look fair? Yes No

Working: _____

Hint. Add the 5 numbers, then divide by 5.

★ Go further

A weekend page. Big challenges. Hints are upside down.

★ Beyond Level 6: Probability of at least one

1 **A coin is flipped 4 times.**

(a) There are ☐ equally likely outcomes.

(b) P(exactly 2 heads) = ☐/16 **(c)** P(at least one tail) = ☐/16

Working: _____

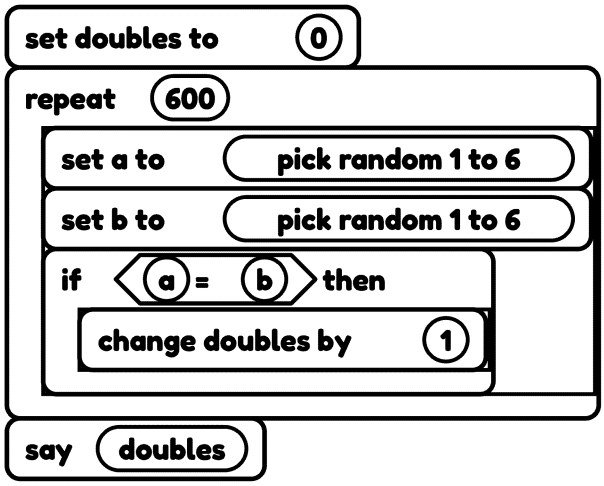
Hint. Which single outcome has no tails? Take it from all 16.

2 **This code rolls 2 dice 600 times. Predict what it says.**

(a) For a given a, b matches it in ☐ out of 6 ways.

(b) It says about ☐.

(c) Change a = b to a + b = 4. It says about ☐.



Hint. List the pairs (a, b) that add to 4.

★ Beyond Level 6: Expected winnings

3 **A game costs 2 dollars to play. You roll a dice and win a prize if you roll a 6. You play 600 times.**

(a) You win about ☐ times.

(b) You pay ☐ dollars. With a 5-dollar prize you win about ☐ dollars.

(c) The game is fair when the prize for a 6 is ☐ dollars.

Working: _____

Hint. Fair means the money won equals the money paid.

✔ Check yourself

No hints. Try each one on your own.

1 A bag has 9 red and 6 blue counters. P(blue) = ☐ / ☐ in simplest form.

Working: _____

2 A dice is rolled 240 times. About ☐ of the rolls are even numbers.

Working: _____

3 Two coins are flipped. P(2 tails) = ☐ / ☐.

Working: _____

4 A spinner has 8 equal parts and 2 are green. In 400 spins about ☐ are green.

Working: _____

My score: ☐ **out of 4. Check the answers at the back.**

What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a helper. Then try Check yourself again.

Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

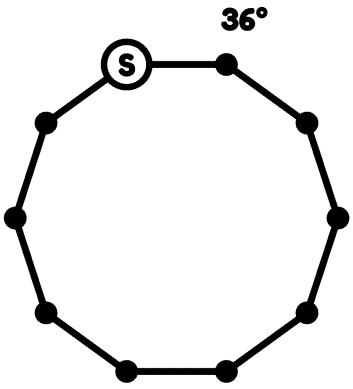
Learn it: Angles in polygons

My goal. I can find the turn angle and the inside angle of a regular polygon. I can predict what a turtle program draws.

A turtle moves forward and turns. Going once round a closed shape, it turns 360° in all and faces the way it began. In a regular polygon with n sides every turn is 360° ÷ n, and the inside angle is 180° – the turn.

Worked example

A turtle draws a regular polygon with 10 sides. Find the turn at each corner and the inside angle.



1. Going once round, the turtle turns 360° in all.
2. There are 10 equal turns: 360° ÷ 10 = 36° at each corner.
3. The turn and the inside angle make a straight line: 180° – 36° = 144°.
4. In code: repeat 10 [move 60, turn ↻ 36].

Answer: The turn is 36° and the inside angle is 144°.

How to find the angles of a regular polygon

1. Going once round, the turtle turns 360° in all.
2. Turn = 360° ÷ number of sides.
3. Inside angle = 180° – turn.
4. Test it in code: repeat (sides) [move, turn].

Your turn

Answer at the back of the book.

Try A regular polygon has 9 sides. The turn at each corner is ☐ ° and the inside angle is ☐ °.

Start here

Do these first. Take your time.

1 **Complete the table for regular polygons.**

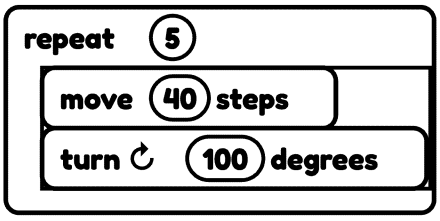
Sides	Turn	Inside angle
5		
12		
	18°	
		172°

2 **Program A: repeat 10 [move 40, turn ↻ 36]. Program B: repeat 12 [move 40, turn ↻ 30]. Program C: repeat 6 [move 40, turn ↻ 30]. One of them draws a regular polygon with inside angles of 150°.**

(a) The program is A B C

(b) Program C turns ☐ ° in all, so its path closes does not close

3 **Trace the code. Write the total turned after each repeat, and the turn from the start direction (the total with every full 360° taken away).**



Repeat	Total	From start
1		
2		
3		
4		
5		

(a) After repeat 5 the turn from the start is ☐ °.

(b) Is the turtle facing the way it started? Yes No

Start here

Keep going. Check your work.

4 **A turtle walks round a closed shape with 5 sides. Its first 4 turns are 60°, 85°, 110° and 50°. It faces the way it started when it is back.**

(a) The last turn is ☐°. **(b)** The inside angle there is ☐°.

(c) The 5 inside angles add up to ☐°.

Working: _____

5 **A regular polygon has 18 sides. Find its angles.**

(a) The turn at each corner is ☐°.

(b) Each inside angle is ☐°.

(c) All the inside angles add up to ☐°.

Working: _____

6 **Circle each turn angle that draws a regular polygon. The number of sides must be a whole number.**

24°	30°	45°	50°	64°	72°	100°	120°	150°

7 **A turtle draws a star with 10 points and winds round the centre 3 times. It runs repeat 10 [move 80, turn ↻ ?].**

(a) The total turning is ☐°.

(b) Each turn is ☐°.

(c) The angle at each point is ☐°.

Working: _____

Start here

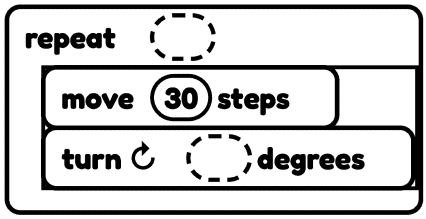
Nearly there. You can do it.

8 **A turtle draws a regular shape with 30 sides. Fill in the blanks.**

(a) repeat ☐

(b) turn ☐ degrees

(c) Each inside angle is ☐°.

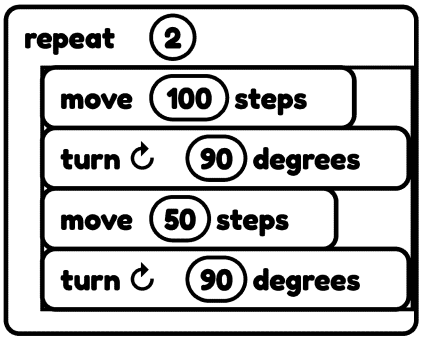


9 **This code draws a rectangle that is not a square. Predict it.**

(a) The turtle turns ☐° in all.

(b) It walks ☐ steps in all.

(c) Change 50 to 80. It walks ☐ steps.



10 **Regular shapes fit around a point when their inside angles add to 360°. Check each group.**

(a) 6 triangles: ☐°

(b) 3 squares: ☐°. Is there a gap? Yes No

(c) 2 hexagons and 2 triangles: ☐°

11 **Each inside angle is a few times the turn angle. Find the number of sides.**

(a) 2 times as big: ☐ sides. (b) 5 times as big: ☐ sides.

(c) 8 times as big: ☐ sides.

Working: _____

Think harder

Hints are upside down. Try first!

1 **A turtle repeats [move 40, turn ↻ 100] over and over. It stops when it first faces the start direction.**

(a) It first faces the start direction after ☐ repeats.

(b) By then it has turned ☐ ° in all. **(c)** That is ☐ full turns.

Working: _____

_____

Hint. Try the total turned after 1, 2, 3 … repeats.

2 **All the inside angles of a regular polygon total 2340°.**

(a) The polygon has ☐ sides. **(b)** Each inside angle is ☐ °.

(c) Each turn is ☐ °.

Working: _____

_____

Hint. What do the inside angles of an n-sided shape add up to?

3 **A triangle, 2 squares and one more regular polygon fit around a point with no gap. All the sides are the same length.**

(a) The inside angle of the fourth shape is ☐ °.

(b) It has ☐ sides.

Working: _____

_____

_____

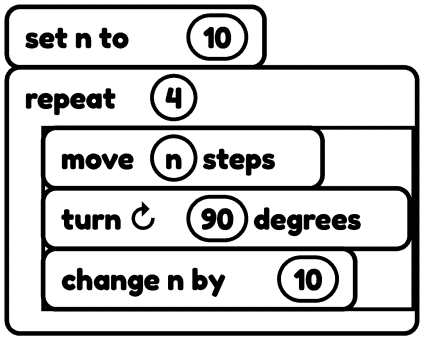
Hint. The angles must add to 360. What is left for the fourth?

★ Go further

A weekend page. Big challenges. Hints are upside down.

★ Beyond Level 6: Where does it end?

1 **The turtle starts at x = 0, y = 0, facing right. A turn ↻ is clockwise. Up means y increases.**



(a) After 4 repeats: x = ☐, y = ☐

(b) After 12 repeats: x = ☐, y = ☐

Hint. Make a table. Write x and y after each repeat.

★ Beyond Level 6: Points of a star

2 **The angle at each point of a star is 180° – the turn. Add up all the point angles.**

(a) 5-point star (turn 144°): the 5 point angles add to ☐°.

(b) 8-point star, 3 times round: the 8 point angles add to ☐°.

(c) 10-point star, 3 times round: each point angle is ☐° and they add to ☐°.

Hint. Find each turn first. Then subtract it from 180 and multiply.

★ Beyond Level 6: A polygon that looks like a circle

3 **The turtle runs repeat 360 [move 1, turn ↻ 1].**

(a) The path is ☐ steps long.

(b) The distance across the shape is about ☐ steps (use 3.14, nearest whole number).

Hint. The path round the shape is about 3.14 times the distance across.

✔ Check yourself

No hints. Try each one on your own.

1 **A regular polygon has 40 sides. Each turn is ☐ ° and each inside angle is ☐ °.**

Working: _____

2 **A turtle turns 10° at each corner of a regular polygon. The polygon has ☐ sides.**

Working: _____

3 **A regular polygon has inside angles of 156° each. It has ☐ sides.**

Working: _____

4 **A turtle goes round a closed 4-sided shape. Its first 3 turns are 95°, 80° and 105°. The last turn is ☐ °.**

Working: _____

My score: ☐ out of 4. Check the answers at the back.

What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a helper. Then try Check yourself again.

Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

Learn it: Clock arithmetic and codes

My goal. I can do clock arithmetic with mod. I can code and decode a secret message with a shift.

On a clock, counting past 12 goes back to 1: 5 o'clock plus 30 hours is 35, and 35 – 24 = 11 o'clock (2 full turns of 12 are 24). n mod m is the remainder when n is divided by m. A Caesar code moves every letter along the alphabet: number the letters A = 0 to Z = 25, add the shift, then take mod 26.

Worked example

Code the word JUMP with a shift of 9.

A	B	C	D	E	F	G	H	I	J	K	L	M
0	1	2	3	4	5	6	7	8	9	10	11	12
N	O	P	Q	R	S	T	U	V	W	X	Y	Z
13	14	15	16	17	18	19	20	21	22	23	24	25

1. **Letter numbers: J = 9, U = 20, M = 12, P = 15.**
2. **Add the shift 9 to each: 18, 29, 21 and 24.**
3. **29 is more than 25, so take mod 26: 29 – 26 = 3. The numbers are 18, 3, 21 and 24.**
4. **Back to letters: 18 = S, 3 = D, 21 = V, 24 = Y.**

Answer: JUMP becomes SDVY.

How to code and decode

1. Write the letter numbers, A = 0 to Z = 25.
2. To code, add the shift. If the total is 26 or more, take away 26.
3. To decode, take away the shift. If the total is below 0, add 26.
4. Change the numbers back into letters.

Your turn

Answer at the back of the book.

Try **Code the word JAB with a shift of 12.** _____

Start here

Do these first. Take your time.

1 **n mod m means: divide n by m and keep the remainder. Find each one.**

(a) 38 mod 7 = ☐ (b) 100 mod 12 = ☐

(c) 59 mod 26 = ☐ (d) 500 mod 26 = ☐

Working: _____

2 **Today is Thursday. A week is a clock with 7 days.**

Mon	Tue	Wed	Thu	Fri	Sat	Sun

(a) 40 days later it is _____

(b) 100 days later it is _____

(c) 20 days ago it was _____

3 **Use the letter table to code each word. Add the shift to every letter number.**

A	B	C	D	E	F	G	H	I	J	K	L	M
0	1	2	3	4	5	6	7	8	9	10	11	12
N	O	P	Q	R	S	T	U	V	W	X	Y	Z
13	14	15	16	17	18	19	20	21	22	23	24	25

(a) HELP with a shift of 3: _____

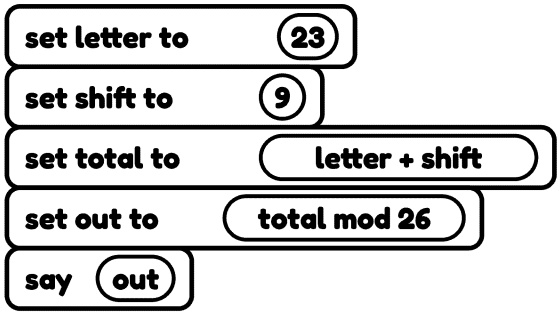
(b) MAZE with a shift of 5: _____

4 **This code changes the number of one letter. Predict what the sprite says.**

(a) It says ☐.

(b) The code letter is ☐.

(c) With letter 17, it says ☐.



Start here

Keep going. Check your work.

5 **Decode each word. Take the shift away from every letter number. If a number is below 0, add 26.**

A	B	C	D	E	F	G	H	I	J	K	L	M
0	1	2	3	4	5	6	7	8	9	10	11	12
N	O	P	Q	R	S	T	U	V	W	X	Y	Z
13	14	15	16	17	18	19	20	21	22	23	24	25

(a) FXAT, shift 9: _____

(b) YNZC, shift 13: _____

6 **The word THE was coded as AOL. Then the next word, XBPJR, was coded with the same shift.**

A	B	C	D	E	F	G	H	I	J	K	L	M
0	1	2	3	4	5	6	7	8	9	10	11	12
N	O	P	Q	R	S	T	U	V	W	X	Y	Z
13	14	15	16	17	18	19	20	21	22	23	24	25

(a) The shift is ☐.

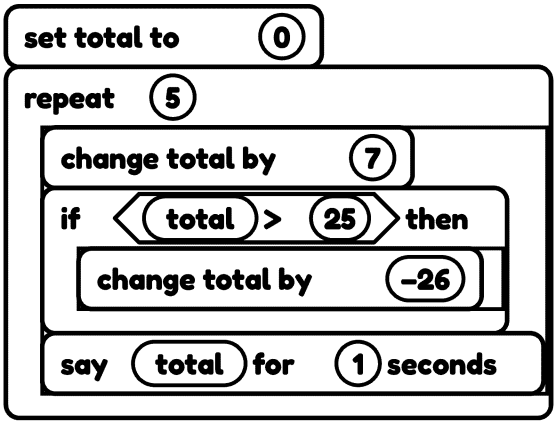
(b) XBPJR decodes to _____

7 **This code adds 7 each time and keeps the number on the 26-letter clock. Predict what it says.**

(a) The 4th number it says is ☐.

(b) The 5th number it says is ☐.

(c) The 3rd number it says is the letter ☐.



Start here

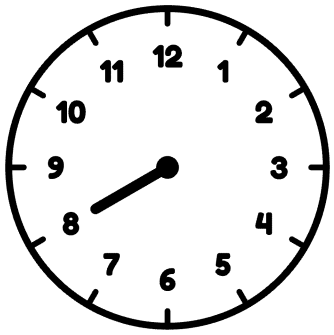
Nearly there. You can do it.

8 **The clock shows 8 o'clock now. A clock goes back to 1 after 12.**

(a) In 7 hours it will be ☐ o'clock.

(b) It is 5 o'clock. 9 hours earlier the clock showed ☐.

(c) At 7 o'clock, 50 hours later it is ☐ o'clock.



9 **Two Caesar shifts can be combined into one shift.**

(a) A shift of 15, then a shift of 18, equals one shift of ☐.

(b) To undo a shift of 7, you can use a shift of ☐.

(c) A shift of 26 makes the message the same reversed

10 **A 24-hour clock goes back to 0 after 23. Write only the hour.**

(a) 22:00 and 7 hours later is ☐:00.

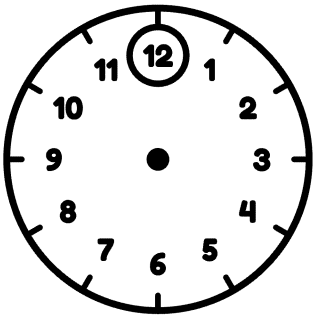
(b) 03:00 and 5 hours earlier is ☐:00.

(c) 09:00 and 100 hours later is ☐:00.

11 **Start at 12 on a clock with 12 numbers. Keep jumping the same number of hours. Count 12 too.**

(a) Jumps of 7 visit ☐ different numbers, including 12.

(b) Jumps of 9 visit ☐ different numbers, including 12.



Think harder

Hints are upside down. Try first!

1 **A number has remainder 3 when divided by 5. It has remainder 4 when divided by 7.**

(a) The smallest such number is ☐.

(b) The next one is ☐.

Working: _____

_____

_____

Hint. List the numbers that leave remainder 3 when divided by 5.

2 **10 January is a Tuesday. This year has 28 days in February.**

(a) From 10 January to 10 April is ☐ days. **(b)** 90 mod 7 = ☐

(c) 10 April is a _____

Working: _____

_____

Hint. January has 31 days, February 28 and March 31.

3 **Mei sends the message QIIX EX RSSR. She says EX is the word AT coded with the same shift.**

(a) The shift is ☐.

(b) The message is _____

Working: _____

_____

Hint. Compare A with its code letter. How far did it move?

★ Go further

A weekend page. Big challenges. Hints are upside down.

★ Beyond Level 6: A code with a changing shift

1 **The 1st letter moves 1, the 2nd moves 2 and the 3rd moves 3. Then the shifts repeat.**

(a) Code MATHS: _____

(b) Decode DQGFU: _____

(c) Why is this code harder to crack than a Caesar code?

the same letter is coded in different ways

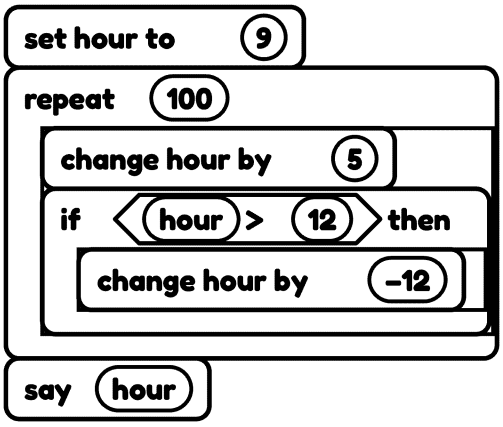
every letter moves the same amount

Hint. Use the shifts 1, 2, 3, 1, 2 in turn, one for each letter.

2 **A sprite goes round a 12-hour clock in jumps of 5 hours. Predict where it ends.**

(a) After 1 jump the hour is ☐.

(b) After 100 jumps the hour is ☐.



Hint. Find 9 + 500, then take away full turns of 12.

★ Beyond Level 6: Check digits

3 **A code of 4 digits ends with a check digit. The 4 digits must add up to a multiple of 10. The check digit is the last digit.**

(a) The first 3 digits are 4, 7, 2. The check digit is ☐.

(b) The first 3 digits are 9, 8, 1. The check digit is ☐.

(c) Is the code 5 3 6 9 valid? Yes No

Working: _____

Hint. What must you add to the first 3 digits to reach a multiple of 10?

✔ Check yourself

No hints. Try each one on your own.

1 It is 9 o'clock. In 40 hours it will be ☐ o'clock.

Working: _____

2 Code NOW with a shift of 6. _____

Working: _____

3 EVQ was coded with a shift of 4. Decode it. _____

Working: _____

4 Today is Friday. In 30 days it will be _____

Working: _____

My score: ☐ **out of 4. Check the answers at the back.**

What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a helper. Then try Check yourself again.

Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

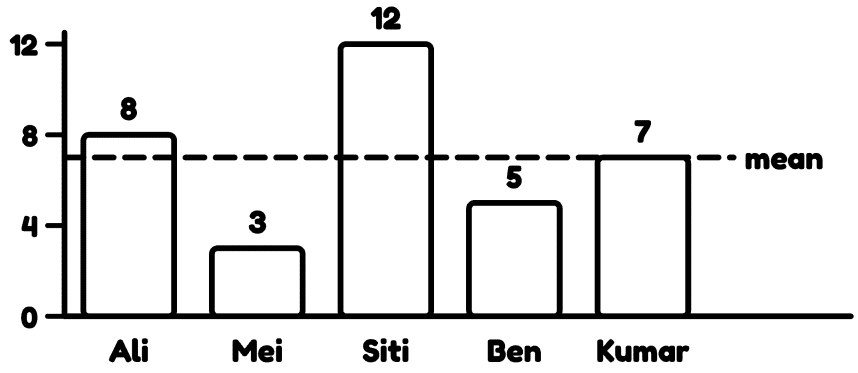
Learn it: Mean, median and mode

My goal. I can find the mean, a missing value from the mean, and compare the mean with the median and mode.

The mean is the total divided by how many values there are. It is the level you get when the towers are made equal. Total = mean × how many. One very large or very small value can pull the mean away from a typical value.

Worked example

Five friends collect shells: Ali 8, Mei 3, Siti 12, Ben 5 and Kumar 7. Find the mean number of shells.



1. **Add all the values: 8 + 3 + 12 + 5 + 7 = 35.**
2. **Divide by how many values there are: 35 ÷ 5 = 7.**
3. **Check by levelling. The towers above 7 have 1 + 5 = 6 extra shells. The towers below 7 are 4 + 2 = 6 short.**

Answer: The mean is 7 shells.

How to work with a mean

1. Add all the values. A value of 0 still counts.
2. Divide the total by how many values there are.
3. Missing value: total = mean × how many, then take away the values you know.
4. Median: put the values in order and take the middle one. With 2 middle values, go halfway between them.
5. Mode: the value that appears most often.

Your turn

Answer at the back of the book.

Try **Kumar saves for 5 weeks. The mean is 10 dollars a week. In the first 4 weeks he saves 8, 14, 12 and 7 dollars. In week 5 he saves ☐ dollars.**

Start here

Do these first. Take your time.

1 **Find the mean of each list.**

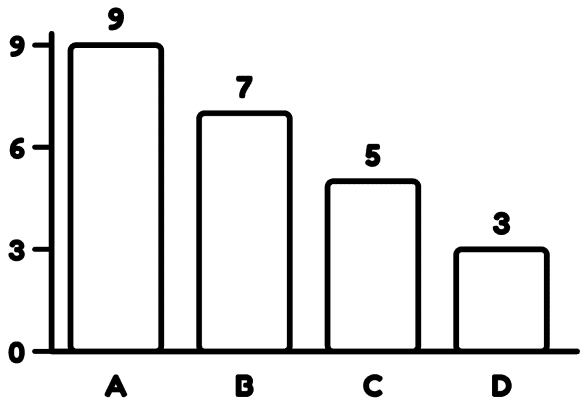
(a) 14, 9, 11, 6: mean = ☐

(b) 9, 0, 6, 0, 10: mean = ☐

(c) 1.5, 2.5, 4, 3: mean = ☐

Working: _____

2 **Move blocks from tall towers to short towers until all towers are equal.**



(a) The mean height is ☐ blocks.

(b) ☐ blocks must move from the tall towers to the short towers.

3 **The mean mass of 5 parcels is 6 kg.**

(a) The 5 parcels weigh ☐ kg in all.

(b) 3 of them weigh 4 kg, 9 kg and 7 kg. The other 2 weigh ☐ kg together.

4 **4 test scores have a mean of 82. Three of the scores are 78, 90 and 85.**

(a) The 4th score is ☐.

(b) A 5th score of ☐ keeps the mean at 82.

Working: _____

Start here

Keep going. Check your work.

5 **The mean of 10, 12 and 14 is 12. Predict, then find what happens when the list changes.**

(a) Add the value 12. The mean goes up stays the same goes down

(b) Add the value 20. The new mean is ☐.

(c) Start again from 10, 12 and 14. Take away the value 14. The new mean is ☐.

6 **The median is the middle value when the values are in order. The mode is the value that appears most often.**

(a) For 12, 7, 9, 7, 5 the median is ☐. **(b)** The mode is ☐.

(c) The mean is ☐. **(d)** For 2, 8, 4, 6 the median is ☐.

7 **A cafe has 7 staff. Their pay in dollars a month is 1500, 1600, 1600, 1700, 1800, 1900 and 9500. The owner earns the 9500.**

(a) The mean pay is ☐ dollars.

(b) The median pay is ☐ dollars.

(c) Which describes what most staff earn better? mean median

8 **The mean mass of 5 children is 40 kg. A 6th child of mass 52 kg joins them.**

(a) The 5 children weigh ☐ kg in all.

(b) With the 6th child the total is ☐ kg.

(c) The new mean is ☐ kg.

Working: _____

Start here

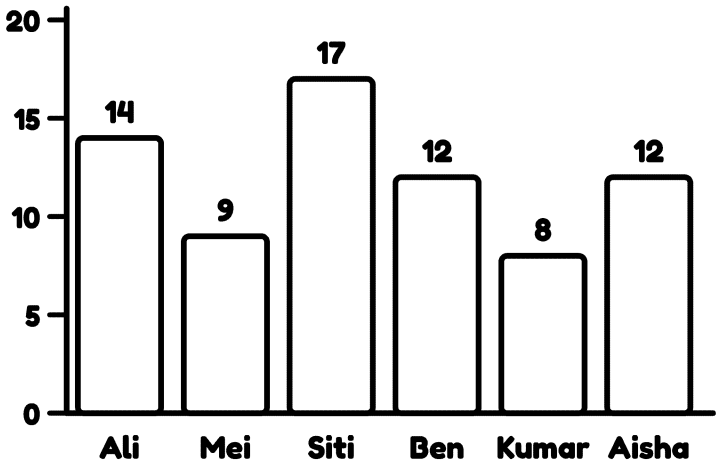
Nearly there. You can do it.

9 **The bar chart shows the number of stickers 6 pupils have.**

(a) They have ☐ stickers in all.

(b) The mean is ☐ stickers.

(c) ☐ pupils have more than the mean.

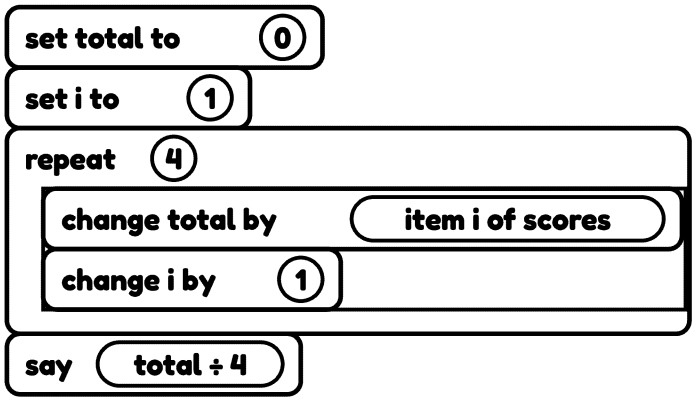


10 **The list scores holds 7, 9, 3 and 5. A loop adds up the items and the program says the mean.**

(a) After the loop, total is ☐.

(b) The program says ☐.

(c) If the last line says total ÷ 3, it says ☐.



11 **The mean of 4, 8 and 12 is 8. Change every number in the list in the same way.**

(a) Add 5 to every number. The new mean is ☐.

(b) Double every number. The new mean is ☐.

Working: _____

Think harder

Hints are upside down. Try first!

1 **Five friends have ages that are whole numbers of years. The mean age is 8 and the median age is 7. The only mode is 5, and the oldest friend is 12.**

(a) The 5 ages add up to ☐.

(b) The 2 youngest friends are each ☐ years old.

(c) The second oldest friend is ☐ years old.

Working: _____

_____

Hint. Put the ages in order. What does the median tell you?

2 **A class has 12 girls and 18 boys. The girls have a mean mass of 40 kg. The boys have a mean mass of 45 kg.**

(a) The girls weigh ☐ kg in all. **(b)** The boys weigh ☐ kg in all.

(c) The mean mass of all 30 is ☐ kg.

(d) It is more than 42.5 because there are more girls boys

Working: _____

_____

Hint. Find each group total first. Then find the total for all 30.

3 **Wei runs 3 laps. The mean lap time is 4.6 minutes. Two laps take 3.9 and 5.4 minutes.**

(a) The 3 laps take ☐ minutes in all.

(b) The third lap takes ☐ minutes.

Working: _____

_____

_____

Hint. Total = mean × how many. Then take away what you know.

★ Go further

A weekend page. Big challenges. Hints are upside down.

★ Beyond Level 6: Pairing up

1 **The numbers 7, 12, 17, 22, ..., 102 go up by 5 each time. There are 20 of them.**

(a) The 1st and the last add up to ☐.

(b) The 20 numbers make ☐ such pairs.

(c) The mean of the 20 numbers is ☐.

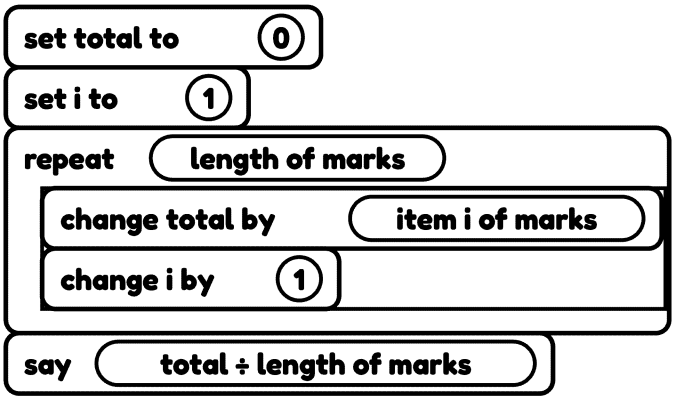
Hint. Pair the 1st with the last, the 2nd with the 2nd last.

2 **The list marks holds 4, 8, 6, 10 and 7. This program adds up the items and says the mean.**

(a) It says ☐.

(b) With repeat 4 and ÷ 5 it says ☐.

(c) With 13 added to the list it says ☐.



Hint. Add up the items the loop reaches. Then divide.

★ Beyond Level 6: A frequency table

3 **In 10 matches a team scored goals as shown.**

Goals	0	1	2	3
Matches	2	4	3	1

(a) The team scored ☐ goals in all.

(b) The mean number of goals is ☐.

(c) The median number of goals is ☐.

Hint. In each column, multiply the goals by the matches.

✔ Check yourself

No hints. Try each one on your own.

1 **The mean of 6, 10, 2 and 14 is ☐.**

Working: _____

2 **For 4, 9, 4, 6, 11 the median is ☐ and the mode is ☐.**

Working: _____

3 **5 numbers have a mean of 6. Four of them are 3, 7, 4 and 9. The fifth is ☐.**

Working: _____

4 **The mean of 4 tests is 70. A 5th test score is 80. The new mean is ☐.**

Working: _____

My score: ☐ out of 4. Check the answers at the back.

What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a helper. Then try Check yourself again.

Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

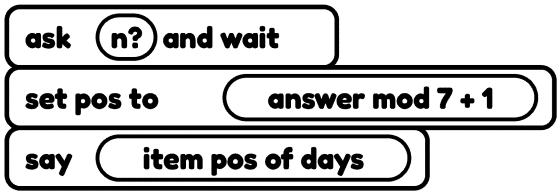
Learn it: Mixed maths and code

My goal. I can use ideas from all the units in a mixed challenge, and plan a small puzzle program: maths first, then blocks, then tests.

A good puzzle program starts with maths. Find the answer by hand first, then turn the rule into blocks, then test it with a number you already know the answer for.

Worked example

Plan a program. It asks for n and says the day n days after Monday. The list days holds Monday to Sunday.



1. **Idea: a week is a clock with 7 days, so we need n mod 7.**
2. **By hand: 10 days after Monday. 10 mod 7 = 3, so the day is Thursday.**
3. **Blocks: a list starts at item 1, so use item (n mod 7 + 1) of days.**
4. **Test with known values: n = 10 gives Thursday, n = 0 and n = 7 give Monday.**

Answer: The program says item 4 of days, Thursday, when n = 10.

How to plan a puzzle program

1. Pick a maths idea and find an answer by hand.
2. Turn the rule into blocks: input, rule, output.
3. Test with a value you know. Also try 0 or 1.
4. Add a twist, then explain why it works.

Your turn

Answer at the back of the book.

Try **The user types 19. What does the program above say?**

_____

Start here

Do these first. Take your time.

1 **A pattern uses 10, 17, 24, 31, … tiles.**

(a) The rule is ☐ × n + ☐. **(b)** The 50th term is ☐.

(c) 164 is term number ☐.

2 **Test numbers for factors. Remember: stop when d × d is more than the number.**

(a) The primes between 90 and 110 are

_____

(b) Write 330 as a product of primes:

_____

3 **Mei's bank balance is –6 dollars, so she owes the bank. She pays in 15 dollars and then spends 20 dollars.**

(a) Her balance is now ☐ dollars.

(b) Later her balance goes from –6 to 21 dollars. It rises by ☐ dollars.

4 **Run the machines backwards.**

(a) A machine does × 5, then – 7. OUT is 43. IN = ☐

(b) OUT = 6n + 1 and OUT is 79. IN = ☐

5 **Think about fair and likely.**

(a) A bag has 5 red and 3 blue counters. P(red) = ☐ / ☐.

(b) A dice is rolled 1200 times. About ☐ of the rolls are 4s.

(c) Ali wins on an even number. Ben wins on 1, 3 or 5. Is it fair? Yes No

Start here

Keep going. Check your work.

6 **Count every possible way.**

(a) A menu has 3 starters, 5 mains and 2 drinks. A meal is one of each: ☐ meals.

(b) 9 friends: every pair shakes hands once: ☐ handshakes.

(c) Cards 2, 4, 6 and 8: ☐ different 3-digit numbers, using no card twice.

7 **Clocks and codes.**

(a) It is 4 o'clock. In 75 hours it is ☐ o'clock.

(b) Code MATH with a shift of 9: _____

8 **Averages.**

(a) The mean of 4 numbers is 15. Three of them are 11, 19 and 14. The 4th is ☐.

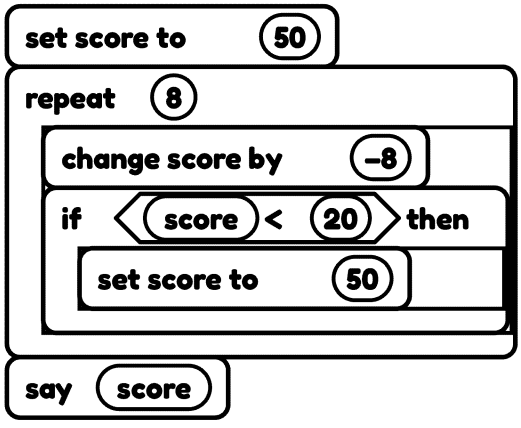
(b) The median of 15, 4, 9, 12 is ☐.

9 **A game starts with a score of 50. This code plays 8 rounds. Run it by hand.**

(a) After round 6 the score is ☐.

(b) The score first resets to 50 after round ☐.

(c) The program says ☐.



Start here

Nearly there. You can do it.

10 **A turtle turns 4° at every corner of a regular polygon.**

(a) The polygon has ☐ sides. **(b)** Each inside angle is ☐ °.

11 **The steps of planning a program are mixed up. A: Test with a value you know. B: Do the maths by hand. C: Build the blocks. D: Pick a maths idea. E: Add a twist.**

The order is _____

12 **A program runs: set n to 11, then repeat 25 times: change n by 6. Then it says n.**

(a) The program says ☐.

(b) That is term number ☐ of 11, 17, 23, ...

(c) The rule for the sequence is ☐ × n + ☐.

13 **Roll 2 dice and multiply the numbers. There are 36 equally likely results.**

(a) ☐ of the 36 products are 12.

(b) P(product is odd) = ☐ / ☐ in simplest form.

(c) The dice are rolled 360 times. About ☐ products are odd.

14 **A sprite starts at x = 15. It repeats 7 times: change x by –4.**

(a) After the 7 repeats, x = ☐.

(b) x is below 0 for the first time after repeat ☐.

Working: _____

Think harder

Hints are upside down. Try first!

1 **A sequence has the rule 6n – 1. Look at its first 10 terms.**

(a) ☐ **of the first 10 terms are prime.**

(b) **The one that is not prime is** ☐.

(c) **Is every term odd?** **Yes** **No**

Terms: _____

_____

Hint. Write out the 10 terms. Test each for factors.

2 **Ali has taken 5 tests. His mean score is 74. After the 6th test his mean is 76.**

(a) **The 5 scores add up to** ☐.

(b) **After the 6th test the total is** ☐. **(c)** **The 6th score is** ☐.

Working: _____

_____

Hint. Total = mean × how many. Do it for 5 tests and for 6.

3 **A 2-digit number has digits that add to 10. Reversing its digits gives another 2-digit number.**

(a) **The primes with digits that add to 10 are** _____

(b) **Those whose reverse is also prime are** ☐ **and** ☐.

Working: _____

_____

_____

Hint. List all numbers with digit sum 10. Cross out the non-primes.

★ Go further

A weekend page. Big challenges. Hints are upside down.

1 A taxi costs 7 dollars for 2 km and 16 dollars for 5 km, by one rule.

```
ask (km?) and wait
say (answer × ? + ?)
```

(a) say (n × ☐ + ☐) **(b)** A 20 km ride costs ☐ dollars.

(c) A fare of 49 dollars is a ride of ☐ km.

Hint. How many km apart are 2 km and 5 km? How much more is the fare?

2 A program spins an 8-part spinner (1 to 8) 800 times and counts the numbers greater than 6.

(a) P(greater than 6) = ☐/8

(b) The program says about ☐.

(c) If the test were greater than 4, it would say about ☐.

```
set count to (0)
repeat (800)
    if (pick random 1 to 8) > (6) then
        change count by (1)
say (count)
```

Hint. Count the numbers that pass the test.

3 This code stops when n reaches 1. Run it by hand.

(a) With n = 6 it says ☐.

(b) With n = 7 it says ☐.

(c) Starting at 7, the largest value n takes is ☐.

```
set steps to (0)
repeat until (n) = (1)
    if (n mod 2) = (0) then
        set n to (n ÷ 2)
    else
        set n to (n × 3 + 1)
    change steps by (1)
say (steps)
```

Hint. Write n after every step.

✔ Check yourself

No hints. Try each one on your own.

1 **4, 11, 18, 25, … The rule is ☐ × n − ☐.**

Working: _____

2 **−7 + 12 − 9 = ☐**

Working: _____

3 **A bag holds 7 red counters and 8 blue counters. P(red) = ☐ / ☐ in simplest form.**

Working: _____

4 **It is 3 o'clock. In 100 hours it will be ☐ o'clock.**

Working: _____

My score: ☐ out of 4. Check the answers at the back.

What next?

- **4 or 3:** well done! Go on to the next unit.
- **2:** read Learn it again. Then try a Start here page once more.
- **1 or 0:** do Learn it and Start here again with a helper. Then try Check yourself again.

Look back

This unit was ☐ easy ☐ just right ☐ hard

The idea I want to remember: _____

Something I want to ask or try: _____

My progress

Tick each box when you finish. Write your Check yourself score.

Unit	Topic	Learn it	Start here	Think harder	Go further	Score	Date
1	Code the pattern	☐	☐	☐	☐	/ 4	
2	Prime hunters	☐	☐	☐	☐	/ 4	
3	Below zero	☐	☐	☐	☐	/ 4	
4	Algebra machines	☐	☐	☐	☐	/ 4	
5	Every possible way	☐	☐	☐	☐	/ 4	
6	Fair or unfair?	☐	☐	☐	☐	/ 4	
7	Turtle geometry	☐	☐	☐	☐	/ 4	
8	Secret codes	☐	☐	☐	☐	/ 4	
9	Data detectives	☐	☐	☐	☐	/ 4	
10	Showcase challenge	☐	☐	☐	☐	/ 4	

Look back

My favourite unit was _____

The hardest idea was _____

Next I want to try _____



Certificate of Completion

Well done,

_____

You finished the Math Inquirers Extension Workbook, Level 6.

_____ _____

Date Signed by a grown-up

Answers

Try each question first. Then check your work here.

Each answer says **Answer** first, then **Why**, and the page of the question. Got one wrong? Read the Why, then try it again. Stuck on the Why? Ask someone to read it with you.

Cover this part of the book with a folded sheet of paper while you work, so you do not see the answers by mistake.

Unit 1 · Code the pattern

Your turn
page 6

Answer: 7, 4, 696
Why: The difference is 7, so compare with 7, 14, 21, 28. Each term is 4 less than the number under it. The rule is 7n − 4. The 100th term is 7 × 100 − 4 = 696.

START HERE

1
page 7

Answer: (a) 17, 5 (b) 5, 2 (c) 202 (d) 18
Why: Pattern 1 has 7 sticks, pattern 2 has 12, pattern 3 has 17: each column adds 5, so the rule is 5n + 2. Pattern 40: 202. For 92: 5n = 90, n = 18.

2
page 7

Answer: (a) 6, 6n + 2, 602 (b) 5, 5n − 2, 498
Why: (a) The difference is 6; each term is 2 more than 6, 12, 18, 24: 6n + 2, and 602. (b) The difference is 5; each term is 2 less than 5, 10, 15, 20: 5n − 2, and 498.

3
page 7

Answer: (a) 21 (b) 297
Why: Loop 1 says 3, then x = 9. Loop 2 says 9, x = 15. Loop 3 says 15, x = 21. Loop 4 says 21, x = 27. The last number said is 21, and x ends as 27. With repeat 50 the last said is 3 + 49 × 6 = 297.

4
page 8

Answer: (a) Yes, term 31 (b) No (c) 302
Why: (a) 157 − 2 = 155 = 5 × 31, so 157 is the 31st term. (b) 205 − 2 = 203 is not a multiple of 5. The terms end in 2 or 7, so 205 is not one. (c) 5 × 60 + 2 = 302.

5
page 8

Answer: (a) 11 (b) 15 (c) 7 (d) 109
Why: The 1st term is 11, so set n to 11. Each jump adds 7. The sprite says a term, then changes n, and 15 terms need 15 says: repeat 15. The last number said is the 15th term: 7 × 15 + 4 = 109.

6
page 8

Answer: (a) 154 (b) 149 (c) 29
Why: The loop adds 5 thirty times: 4 + 5 × 30 = 154. That is the 31st term. The 30th term is 5 × 30 − 1 = 149. From the 1st to the 30th term there are only 29 jumps, so use repeat 29.

7
page 8

Answer: (a) 5n − 1 (b) 3n + 7 (c) 8n − 5
Why: Put n = 1 and n = 2 into each rule. (a) 5 − 1 = 4 and 10 − 1 = 9. (b) 3 + 7 = 10 and 6 + 7 = 13. (c) 8 − 5 = 3 and 16 − 5 = 11. The other rules fail on the 1st term.

8
page 9

Answer: (a) 36 (b) 7, 8 (c) 218 (d) 28
Why: Each day adds $7, so the rule is 7n + 8. Day 4: 36. Day 30: 7 × 30 + 8 = 218. 7n + 8 > 200 needs 7n > 192: 7 × 27 = 189 is too small and 7 × 28 = 196 works, so Day 28 ($204).

9
page 9

Answer: (a) 59 (b) 41 (c) 17
Why: The 1st term needs 0 repeats and each repeat adds 1 to the term number. (a) 60 − 1 = 59 repeats. (b) 1 + 40 = the 41st term. (c) 25 − 8 = 17 repeats. The repeats count the jumps, not the terms.

10 page 9

Answer: (a) 4, 7, 10 (b) 25 (c) 4, 1
Why: (a) n = 1, 2, 3 give 3 × 1 + 1 = 4, 7, 10. (b) The 8th number is 3 × 8 + 1 = 25. (c) 5, 9, 13 go up by 4, so use 4 × n + 1: 4 × 1 + 1 = 5.

11 page 9

Answer: (a) No (b) 6, 2 (c) 6, 4
Why: The rule 6n gives 6 for the 1st term, but the 1st term is 2. Each term is 4 less than 6n: 6 − 4 = 2, 12 − 4 = 8. The difference only tells you the times table. You must still compare with the terms.

THINK HARDER

1 page 10

Answer: (a) 63 (b) 13 (c) 12
Why: (a) The terms go down by 5: 5 × 1 = 5 and 58 + 5 = 63, so 63 − 5n. (b) 63 − 50 = 13. (c) 5 × 12 = 60 gives 3, and 5 × 13 = 65 would be below 0: term 12.

2 page 10

Answer: (a) 21 (b) 20 (c) 86
Why: The 5th term is 4 × 5 + 2 = 22. Q adds 4 sixteen times: 22 + 64 = 86, which is term 5 + 16 = 21. P starts at the 1st term, so it needs 21 − 1 = 20 repeats (6 + 20 × 4 = 86).

3 page 10

Answer: (a) 5 (b) 6 (c) 5 (d) 6, 1
Why: From the 4th to the 9th term is 5 jumps and the terms rose by 30, so each jump is 6. Back 3 jumps from 23: 5. The rule is 6n − 1.

GO FURTHER

1 page 11

Answer: (a) 10, 18 (b) 28, 40 (c) 130
Why: The extra sticks are 6, 8, then 10 and 12 (they go up by 2): 18 + 10 = 28 and 28 + 12 = 40. Staircase 10: 4 + 6 + 8 + ... + 22 = 130. (The rule is n × n + 3n.)

2 page 11

Answer: (a) 4, 5 (b) 31 (c) 65
Why: Places 1, 3, 5, ... say 3, 7, 11, ...: up by 4. Places 2, 4, 6, ... say 20, 25, 30, ...: up by 5. Place 15 is the 8th of the first kind: 3 + 7 × 4 = 31. Place 20 is the 10th of the second: 20 + 9 × 5 = 65.

3 page 11

Answer: (a) 14 (b) 105
Why: t after each repeat: 1, 3, 6, 10, 15, 21, 28, 36, 45, 55, 66, 78, 91, 105. It stops when t is above 100, after repeat 14: k = 14, t = 105. (Triangle numbers.)

CHECK YOURSELF

1 page 12

Answer: 7, 2, 702
Why: The difference is 7. Each term is 2 more than 7, 14, 21, 28, so the rule is 7n + 2. The 100th term is 7 × 100 + 2 = 702.

2 page 12

Answer: No
Why: The rule is 5n + 2. 125 − 2 = 123 is not a multiple of 5. The terms end in 2 or 7, so 125 is not a term.

3 page 12
Answer: 128, term 21
Why: 8 + 20 × 6 = 128. From the 1st term to the 21st term there are 20 jumps, so 128 is the 21st term. Check: 6 × 21 + 2 = 128.

4 page 12
Answer: 8
Why: The code changes n before it says it. 5 + 3 = 8, so the first number said is 8, not the 1st term 5. To say 5 first, say n before changing it.

Unit 2 · Prime hunters

Your turn page 13
Answer: No, 11 × 13
Why: Test 2, 3, 5 and 7: each leaves a remainder (143 mod 7 = 3). Then test 11, because 11 × 11 = 121 is less than 143: 11 × 13 = 143. So 143 is not prime.

START HERE

1 page 14
Answer: (a) 1, 2, 3, 6, 7, 14, 21, 42 (b) 9 (c) 10
Why: 42: 1 × 42, 2 × 21, 3 × 14, 6 × 7, so 8 factors. 100: 1 × 100, 2 × 50, 4 × 25, 5 × 20 and 10 × 10. The pairs give 8 factors and 10 pairs with itself, so 9 in all.

2 page 14
Answer: 53, 59, 67, 73, 79
Why: Test each number. 51 = 3 × 17, 57 = 3 × 19, 63 = 7 × 9, 69 = 3 × 23 and 77 = 7 × 11 are not prime. The primes are 53, 59, 67, 73 and 79. They have no factor except 1 and themselves.

3 page 14
Answer: (a) 2 × 2 × 3 × 7 (b) 2 × 3 × 3 × 5
Why: 84 = 4 × 21 = 2 × 2 × 3 × 7. 90 = 9 × 10 = 3 × 3 × 2 × 5, so 90 = 2 × 3 × 3 × 5.

4 page 14
Answer: 3 + 23, 7 + 19, 13 + 13
Why: 26 − 3 = 23, 26 − 7 = 19 and 26 − 13 = 13 are all prime. 26 − 2, 26 − 5, 26 − 11 and 26 − 17 are 24, 21, 15, 9: not prime. 3 ways.

5 page 15
Answer: (a) 4 (b) 20
Why: n mod d for d = 1 to 6 is 0, 0, 2, 0, 0, 2, so count goes 1, 2, 2, 3, 4, 4: it says 4. But 20 has 6 factors (1, 2, 4, 5, 10, 20): the loop stops at d = 6 and misses 10 and 20. Repeat 20 times to count all.

6 page 15
Answer: (a) 53 (b) 71
Why: (a) Primes from 40 to 60: 41, 43, 47, 53, 59. Only 53 has digits adding to 8. (b) Primes from 60 to 80: 61, 67, 71, 73, 79. Only 71 + 2 = 73 is prime.

7 page 15
Answer: (a) 5 (b) 9 (c) 64
Why: (a) 81: 1 × 81, 3 × 27 and 9 × 9 give 1, 3, 9, 27, 81. (b) 9 is paired with itself. (c) 64 = 8 × 8 is a square, so it has an odd number (7). 50 has 6 factors and 72 has 12.

8 page 16
Answer: (a) Yes (b) Yes (c) No
Why: (a) 72 is in the 4 times table (4 × 18): Yes. (b) 2 + 4 + 5 + 7 = 18, a multiple of 3: Yes. (c) 6 + 8 + 3 + 5 = 22, not a multiple of 9: No.

9 page 16

Answer: (a) 1, 2, 3, 1 (b) Yes (c) No, 7 × 17
Why: 113 leaves remainders 1, 2, 3 and 1 when divided by 2, 3, 5 and 7. None is 0, and 11 × 11 = 121 is too big to need testing, so 113 is prime. 119 = 7 × 17 is not prime.

10 page 16

Answer: (a) 14 (b) 5 (c) 70
Why: x goes 7, 14, 21, 28, 35, 42, 49, 56, 63, 70. The code says x only when x mod 2 = 0, that is when x is even. It says 14, 28, 42, 56 and 70. The first is 14, there are 5 numbers, and the last is 70.

11 page 16

Answer: (a) 2, 3, 4, 6, 9, 12, 18 (b) 7
Why: The number of rows must be a factor of 36 other than 1 and 36: 2, 3, 4, 6, 9, 12 and 18. That is 7 ways. (2 rows of 18 and 18 rows of 2 count as different ways.)

THINK HARDER

1 page 17

Answer: (a) 16 (b) 36 (c) 48
Why: (a) 16 has 1, 2, 4, 8, 16: 5 factors, and no smaller number has 5. (b) 36 has 1, 2, 3, 4, 6, 9, 12, 18, 36: 9 factors. (c) 48 has 1, 2, 3, 4, 6, 8, 12, 16, 24, 48: 10 factors. Numbers below 48 have fewer.

2 page 17

Answer: (a) 42, 66, 78 (b) 8 (c) 66
Why: p = 7, 11 or 13 gives 42, 66 and 78 (p = 17 gives 102, too big). 42 has the factors 1, 2, 3, 6, 7, 14, 21, 42: 8 factors, and so do 66 and 78.

3 page 17

Answer: (a) 101, 103, 107, 109, 113 (b) 2 (c) 105 and 119
Why: Test 2, 3, 5 and 7 only: 11 × 11 = 121 is above 120. 101, 103, 107, 109 and 113 have no such factor. The odd non-primes are 105, 111, 115, 117 and 119. Differ by 2: 101 and 103, 107 and 109. (c) 105 = 7 × 15 and 119 = 7 × 17.

GO FURTHER

1 page 18

Answer: (a) 12 (b) 18 (c) 60
Why: (a) 72: 1, 2, 3, 4, 6, 8, 9, 12, 18, 24, 36, 72 is 12 factors. In 2 × 2 × 2 × 3 × 3 a factor uses 0 to 3 twos (4 choices) and 0 to 2 threes (3 choices): 4 × 3 = 12. (b) 180 = 2 × 2 × 3 × 3 × 5: 3 × 3 × 2 = 18 factors. (c) 60 = 2 × 2 × 3 × 5 has 3 × 2 × 2 = 12 factors. Smaller numbers have fewer.

2 page 18

Answer: (a) 17 (b) 29
Why: Cross out multiples of 2, 3, 5 and 7 (11 × 11 = 121 is above 60, so stop at 7). Left: 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59: 17 primes. The 10th is 29.

3 page 18

Answer: (a) 13 (b) 11 (c) 97
Why: n = 221: d goes 2, 3, ... and stops at 13 (221 mod 13 = 0), after 11 repeats. n = 97: d goes on to 97, because 97 is prime. If it says n itself, n is prime.

CHECK YOURSELF

1 page 19
Answer: 1, 2, 4, 5, 8, 10, 20, 40
Why: 40 = 1 × 40 = 2 × 20 = 4 × 10 = 5 × 8. The factors are 1, 2, 4, 5, 8, 10, 20 and 40.

2 page 19
Answer: No
Why: The digits add to 1 + 4 + 7 = 12, a multiple of 3, so 3 divides 147 (147 = 3 × 49). It is not prime.

3 page 19
Answer: 2 × 2 × 5 × 7
Why: 140 = 2 × 70 = 2 × 2 × 35 = 2 × 2 × 5 × 7. Every number here is prime.

4 page 19
Answer: 5, 4
Why: 47 = 7 × 6 + 5, so the remainder is 5. 100 = 12 × 8 + 4, so the remainder is 4.

Unit 3 · Below zero

Your turn page 20
Answer: −11
Why: Falling moves left. From −3 jump left 8 steps: −3 + (−8) = −11. Check: from −11 jump right 8 to reach −3.

START HERE

1 page 21
Answer: (a) −11, −6, −3, 0, 2, 8 (b) >, >, <
Why: The smallest number is the furthest left on the number line: −11, −6, −3, 0, 2, 8. (b) On the number line, the number on the right is greater. −4 is to the right of −10, 0 is right of −2, and −15 is left of 1.

2 page 21
Answer: (a) −9, −3, 5 (b) 6, 8
Why: A is 1 step right of −10: −9. B is 1 step left of −2: −3. C is 1 step left of 6: 5. From −9 to −3 is 6 steps. From −3 to 5 is 3 + 5 = 8 steps.

3 page 21
Answer: (a) −6 (b) 4 (c) −3 (d) 10 (e) 7
Why: Read each mark: 6 a.m. is −6 °C, noon is 4 °C and 9 p.m. is −3 °C. From −6 up to 4 is 6 steps to 0 and 4 more: 10 degrees. From 4 down to −3 is 4 steps to 0 and 3 more: 7 degrees.

4 page 22
Answer: (a) 4 (b) −6 (c) −12 (d) 11
Why: (a) From −6 jump right 10: 4 steps to 0, then 6 more: 4. (b) From 3 jump left 9: 3 steps to 0, then 6 more: −6. (c) From −7 jump left 5: −12. (d) From −4 jump right 15: 4 steps to 0, then 11 more: 11.

5 page 22
Answer: (a) 11 (b) 7 (c) −6 (d) −4
Why: (a) From −5 to 6 is 5 steps to 0 and 6 more: 11. (b) 4 − 7 = −3, so 7. (c) −6 + 8 = 2, so the missing number is −6. (d) −4 − 5 = −9, so the missing number is −4.

6 page 22
Answer: (a) 13 (b) 5 (c) 45
Why: (a) From −9 to 4: 9 steps to 0 and 4 more is 13. (b) From −6 to −1: 5 steps, no need to cross 0. (c) From −14 to 31: 14 steps to 0 and 31 more is 45.

7 page 22
Answer: (a) −25 (b) −60
Why: (a) 4 equal moves must go from 100 down to 0. That is 100 left in 4 moves, so each move is −25. (b) 8 moves of −20 make −160, and 100 − 160 = −60.

8 page 22
Answer: (a) −3 (b) −8
Why: (a) The distances from 0 are 9, 5, 3, 7 and 4. −3 is only 3 steps from 0. (b) −8 is 8 steps from 0 and 6 is 6 steps from 0, so −8 is further.

9 page 23
Answer: (a) −15 (b) −65 (c) 3
Why: x starts at 60. Loop 1: 60 − 25 = 35. Loop 2: 10. Loop 3: −15. Loop 4: −40. Loop 5: −65. The 3rd number said is −15. The last is −65. The numbers below 0 are −15, −40 and −65: 3 of them.

10 page 23
Answer: (a) −15 (b) 15 (c) 35
Why: (a) 25 − 40 = −15. (b) −15 + 30 = 15. (c) From −15 up to 20: 15 dollars to reach 0, then 20 more, 35 dollars.

11 page 23
Answer: (a) 83 (b) 29 (c) 60
Why: (a) From −35 up to 0 is 35, then 48 more: 83 m. (b) From −35 up to −6 is 29 m. (c) 48 + 12 = 60 m.

THINK HARDER

1 page 24
Answer: (a) 10 (b) 3 (c) 4
Why: (a) 6 − (−4): from −4 to 6 is 10 steps. (b) −2 − (−5): from −5 to −2 is 3 steps. (c) In both, subtracting a negative number gave the same answer as adding the matching positive number: 6 + 4 and −2 + 5.

2 page 24
Answer: (a) 6 (b) 60
Why: Each repeat changes x by 35 − 10 = 25. After the repeats x is −65, −40, −15, 10, 35, 60. After 5 repeats x = 35, which is not greater than 40, so it repeats a 6th time. It stops with x = 60.

3 page 24
Answer: (a) −8 (b) −2 (c) 7
Why: Tuesday: 3 − 11 = −8. Wednesday: −8 + 6 = −2. Thursday is −4, which is 2 below Wednesday. From Monday (3) to Thursday (−4): 3 steps to 0 and 4 more, so it fell 7 degrees.

GO FURTHER

1 page 25
Answer: (a) −24, −22 (b) 3 (c) 22
Why: After 3 minutes A is at −24 and B at −22, so B is above A (after 2 minutes A −20 and B −28). After 5 minutes A −32 and B −10: 22 m apart.

2 page 25
Answer: (a) 13 (b) 6 (c) 0
Why: It says how far apart a and b are. a = −8, b = 5: a is not greater, so 5 − (−8) = 13. a = −3, b = −9: a is greater, so −3 − (−9) = 6. Equal numbers give 0.

3 page 25
Answer: (a) 4 (b) 6 (c) 21
Why: (a) 5, 4, 3, 2, 1 BC: 4 steps. (b) 3 BC, 2 BC, 1 BC, AD 1, 2, 3, 4: 6 steps. (c) 10 BC to 1 BC is 9 steps, 1 more to AD 1, then 11 to AD 12: 21 years.

CHECK YOURSELF

1 page 26
Answer: 5, −7
Why: From −9 jump right 14: 9 steps to 0 and 5 more is 5. From 6 jump left 13: 6 steps to 0 and 7 more is −7.

2 page 26
Answer: −7, −2, 0, 4
Why: On a number line, from left to right: −7, −2, 0, 4.

3 page 26
Answer: −3
Why: −6 + 11 = 5. Then 5 − 8 = −3.

4 page 26
Answer: −25
Why: 3 moves of −15 make −45. 20 − 45 = −25.

Unit 4 · Algebra machines

Your turn page 27
Answer: 8
Why: Undo the last step first: 41 − 9 = 32. Then undo × 4: 32 ÷ 4 = 8. Check: 8 × 4 + 9 = 41.

START HERE

1 page 28
Answer: (a) 3, 2 (b) 32 (c) 16
Why: IN goes up 2 and OUT goes up 6, so each 1 of IN adds 3: × 3. 3 × 1 = 3 but OUT is 5, so add 2: OUT = 3n + 2. IN 10 gives 32. For OUT 50: 3n = 48, n = 16.

2 page 28
Answer: (a) 3, 5 (b) 55 (c) 35
Why: IN goes up 2 and OUT goes up 6, so × 3. 3 × 2 = 6 but OUT is 1, so take away 5: OUT = 3n − 5. IN 20 gives 55. For OUT 100: 3n = 105, n = 35.

3 page 28
Answer: (a) 0 (b) 73 (c) 1 (d) 11
Why: Start at 0 and add 1 until 7 × n − 4 is 73. 7 × 10 − 4 = 66 is too small, and 7 × 11 − 4 = 73, so it says 11. Check: (73 + 4) ÷ 7 = 11.

4 page 29
Answer: (a) 20, 26 (b) 26, 32 (c) No
Why: IN 7: P gives 7 × 2 = 14, then 20. Q gives 7 + 6 = 13, then 26. IN 10: P gives 26 and Q gives 32. The order of the steps matters, so P and Q do not give the same OUT.

5 page 29
Answer: (a) 15 (b) 4 (c) 31
Why: IN 5 times 3 is 15, so the middle box is 15. 15 + 4 = 19, so the last step is add 4. IN 9: 9 × 3 = 27, then 27 + 4 = 31.

6 page 29
Answer: (a) 33 (b) 97 (c) 16
Why: out = answer × 4 − 3. Type 9: 36 − 3 = 33. Type 25: 100 − 3 = 97. For 61, run it backwards: 61 + 3 = 64, then 64 ÷ 4 = 16.

7 page 29
Answer: (a) 13, 24 (b) 45 (c) 5, 5
Why: Kumar has n + 5 and Jenny has 3 × n. When n = 8: Kumar has 13, Jenny has 24, and all 3 together have 8 + 13 + 24 = 45. In general n + (n + 5) + 3n = 5n + 5.

8 page 30
Answer: (a) 11 (b) 9 (c) 7 (d) 40
Why: (a) 63 − 8 = 55, 55 ÷ 5 = 11. (b) 74 + 7 = 81, 81 ÷ 9 = 9. (c) 66 ÷ 6 = 11, 11 − 4 = 7. (d) 11 − 3 = 8, 8 × 5 = 40.

9 page 30
Answer: (a) 27 (b) 3, 12 (c) 16
Why: Cost = 3 × n + 12. 5 extra GB cost 3 × 5 + 12 = 27 dollars. For 60 dollars: 3n + 12 = 60, so 3n = 48 and n = 16.

10 page 30
Answer: (a) 1 (b) 1 (c) 1
Why: Start with 7: 11, 33, 24, 8, then 8 − 7 = 1. Start with 12: 16, 48, 39, 13, then 1. With n: n + 4, 3n + 12, 3n + 3, n + 1, then n + 1 − n = 1. Always 1.

11 page 30
Answer: (a) 12 (b) 5 (c) 4 and 18
Why: 6 is not more than 10, so it says 12. 15 is more than 10, so it says 5. Says 8: answer − 10 = 8 gives 18, answer × 2 = 8 gives 4.

THINK HARDER

1 page 31
Answer: (a) 35 (b) 12 (c) 6, 5
Why: IN 5: X gives 13, Y gives 39 − 4 = 35. For OUT 77, undo Y: (77 + 4) ÷ 3 = 27, then undo X: (27 − 3) ÷ 2 = 12. Joined rule: 3(2n + 3) − 4 = 6n + 5. Check: 6 × 12 + 5 = 77.

2 page 31
Answer: (a) 5, 4 (b) 12 (c) 36
Why: Ali n, brother n + 4, mother 3n: 5n + 4 = 64, so n = 12. Ali is 12, his brother 16, their mother 36 (12 + 16 + 36 = 64).

3 page 31
Answer: (a) 25 (b) 5, 2 (c) 20
Why: IN up 5 and OUT up 25, so × 5. 5 × 3 = 15 but OUT is 13, so take away 2: OUT = 5n − 2. For OUT 98: 5n = 100, n = 20.

GO FURTHER

1 page 32
Answer: (a) 9, 17 (b) 6, 25
Why: IN 2: A gives 9 and B gives 17, so B is ahead by 8. A gains 2 for every 1 of IN, so the lead is gone in 4 steps, at IN 6. Both give 25. As an equation, 4n + 1 = 2n + 13 gives 2n = 12.

2 page 32
Answer: (a) 31 (b) 21 (c) No
Why: (a) n + (n + 1) + (n + 2) + (n + 3) = 4n + 6 = 130, so n = 31. (b) 5 odd numbers: n + (n + 2) + (n + 4) + (n + 6) + (n + 8) = 5n + 20 = 125, so n = 21. (c) 4n + 6 = 100 gives n = 23.5, not a whole number: No.

3 page 32
Answer: (a) 18 (b) 123 (c) 19
Why: 7n − 3 > 120 needs 7n > 123, so n > 17.57: n = 18 gives 123. It says 18, and OUT is 123. If 120 changes to 123, then 123 is not greater than 123, so it keeps going to n = 19.

CHECK YOURSELF

1 page 33
Answer: 5, 1
Why: OUT goes up 5 for each 1 of IN, so × 5. 5 × 1 = 5, and OUT is 6, so add 1: OUT = 5n + 1.

2 page 33
Answer: 6
Why: Undo − 5: 49 + 5 = 54. Undo × 9: 54 ÷ 9 = 6. Check: 6 × 9 − 5 = 49.

3 page 33
Answer: 12
Why: 57 − 9 = 48. 48 ÷ 4 = 12. Check: 4 × 12 + 9 = 57.

4 page 33
Answer: 13
Why: n + (n + 4) = 2n + 4 = 30. So 2n = 26 and n = 13. Check: 13 + 17 = 30.

Unit 5 · Every possible way

Your turn page 34
Answer: 24
Why: Choices one after another multiply: 4 × 6 = 24. Check: each of the 4 crusts goes with 6 toppings, which makes 4 groups of 6.

START HERE

1 page 35
Answer: (a) 30 (b) 36 (c) 6
Why: Choices one after another multiply: 5 × 3 × 2 = 30. With 6 frames: 6 × 3 × 2 = 36. A red frame is 1 frame choice: 1 × 3 × 2 = 6.

2 page 35
Answer: (a) 12 (b) 2 (c) 3
Why: The coin has 2 results and the dice has 6, so 2 × 6 = 12 results, one for each cell. Tails with more than 4: T5 and T6, so 2. Heads with an even number: H2, H4 and H6, so 3.

3 page 35
Answer: (a) 10
Why: A corner is reached from the corner on its left or the corner below, so add those 2 numbers. Middle row: 2, 3, 4. Top row: 3, 6, 10. There are 10 routes to End.

4 page 36
Answer: (a) 28 (b) 55 (c) 12
Why: Each new friend shakes hands with all the earlier friends. 6 friends: 1 + 2 + 3 + 4 + 5 = 15. 8 friends: 1 + … + 7 = 28. 36 = 1 + … + 8, so 9 friends.

5 page 36
Answer: (a) 3 (b) 15 (c) 18
Why: A is said once for each turn of the outer repeat: 3 times. B is said 5 times in each turn: 3 × 5 = 15 times. In all it says 3 + 15 = 18 things.

6 page 36
Answer: (a) 64 (b) 24 (c) 16
Why: Letters may repeat: 4 × 4 × 4 = 64. No repeats: 4 choices, then 3, then 2: 4 × 3 × 2 = 24. First letter A and repeats allowed: 1 × 4 × 4 = 16.

7 page 36
Answer: (a) 28 (b) 20 (c) 35
Why: Every pair of corners gives a line: 8 × 7 ÷ 2 = 28. 8 of the lines are the sides, so 28 − 8 = 20 are diagonals. 10 corners: 10 × 9 ÷ 2 = 45 lines, 10 are sides, so 35 diagonals.

8 page 37
Answer: (a) 60 (b) 12
Why: Gold has 5 choices, then silver 4, then bronze 3: 5 × 4 × 3 = 60. If Ali wins gold, silver has 4 choices and bronze 3: 4 × 3 = 12.

9 page 37

Answer: (a) 21 (b) 6 (c) 12
Why: For each a the inner repeat says 2 numbers: 11, 12, then 21, 22, then 31, 32. The 3rd is 21 and it says 6 numbers. With 4 inner repeats it says 3 × 4 = 12 numbers.

10 page 37

Answer: (a) 18 (b) 8
Why: First digit: 3 choices (not 0), second: 3 (0 is allowed), third: 2. 3 × 3 × 2 = 18. Odd numbers end in 1 or 3; then the first digit has 2 choices and the middle digit 2: 2 × 2 × 2 = 8.

11 page 37

Answer: (a) 15 (b) 28
Why: Row of 5: 5 + 4 + 3 + 2 + 1 = 15 (5 of length 1, 4 of length 2, ...). Row of 7: 7 + 6 + 5 + 4 + 3 + 2 + 1 = 28.

THINK HARDER

1 page 38

Answer: (a) 3, 6 (b) 18
Why: Routes to X: right 2 and up 1, so 3 routes (RRU, RUR, URR). Routes from X to End: right 2 and up 2, so 6 routes. Every route through X is one of the 3 followed by one of the 6: 3 × 6 = 18.

2 page 38

Answer: (a) 24 (b) 12 (c) 6
Why: 4 × 3 × 2 × 1 = 24 numbers. Even numbers end in 2 or 4: 2 × 3 × 2 × 1 = 12. Multiples of 4 end in 12, 24 or 32, and each leaves 2 numbers: 3 × 2 = 6.

3 page 38

Answer: (a) 6 (b) 10 (c) 28
Why: Each pair of points on the line makes one triangle with P. 4 points: 4 × 3 ÷ 2 = 6 pairs. 5 points: 5 × 4 ÷ 2 = 10. 8 points: 8 × 7 ÷ 2 = 28. This is the handshake count again.

GO FURTHER

1 page 39

Answer: (a) 125 (b) 64 (c) 61
Why: All codes: 5 × 5 × 5 = 125. Codes with no 5 use only 1, 2, 3 or 4: 4 × 4 × 4 = 64. Every other code has at least one 5: 125 − 64 = 61.

2 page 39

Answer: (a) 16 (b) 12 (c) 27
Why: Each repeat runs 4 times: 4 × 4 = 16 pairs. a × b is odd only when both are odd: 2 × 2 = 4 pairs. So 16 − 4 = 12 even products. With 6 and 6: 36 pairs, 9 odd, 27 even.

3 page 39

Answer: (a) 3 (b) 5 (c) 13 (d) 34
Why: The last move onto step n is 1 step from step n − 1 or 2 steps from step n − 2, so ways(n) = ways(n − 1) + ways(n − 2). For 1 to 8 steps: 1, 2, 3, 5, 8, 13, 21, 34.

CHECK YOURSELF

1 page 40

Answer: 15
Why: Choices one after another multiply: 5 × 3 = 15.

2 page 40
Answer: 20
Why: Each of the 5 teams plays 4 other teams at home: 5 × 4 = 20 matches. (Each pair is counted once for home and once for away.)

3 page 40
Answer: 6
Why: 3 choices, then 2, then 1: 3 × 2 × 1 = 6 (258, 285, 528, 582, 825, 852).

4 page 40
Answer: 100
Why: 5 × 5 × 4 = 100.

Unit 6 · Fair or unfair?

Your turn page 41
Answer: 1/4
Why: 20 counters in all and 5 are white: 5/20. Divide the top and bottom by 5: 1/4.

START HERE

1 page 42
Answer: (a) 1/5 (b) 1/2 (c) 150
Why: There are 10 equally likely parts. Green: 2 parts, so 2/10 = 1/5. Not red: 3 blue + 2 green = 5 parts, so 5/10 = 1/2. Blue is 3/10, and 3/10 × 500 = 150.

2 page 42
Answer: (a) 6 (b) 6 (c) 1/3
Why: The coin has 2 results and the spinner has 3, so 2 × 3 = 6 equally likely results, one for each cell. HA is 1 of the 6, so 1/6. Tails with B or C: TB and TC, so 2/6 = 1/3.

3 page 42
Answer: (a) 1/6 (b) 17 (c) random rolls
Why: One roll is 6 with probability 1/6. In 100 rolls we expect 1/6 × 100 = 16.7, so about 17. The answers differ because every roll is random, so each run gives a slightly different count.

4 page 43
Answer: (a) 1/3 (b) 2/3 (c) 9/20
Why: The bag has 4 + 5 + 6 = 15 counters. Blue: 5/15 = 1/3. Red or green: 4 + 6 = 10, so 10/15 = 2/3. With 5 more red counters there are 9 red out of 20: 9/20.

5 page 43
Answer: (a) 4 (b) 8 (c) No
Why: The primes up to 10 are 2, 3, 5, 7: Ali wins on 4 numbers and Ben on the other 6. 4 is not 6, so it is not fair: Ben is favoured.

6 page 43
Answer: (a) 50 (b) 100 (c) 45 (d) No
Why: (a) 1/6 × 300 = 50. (b) 5 or 6 is 2 out of 6, so 1/3 × 300 = 100. (c) 1/2 × 90 = 45. (d) 54 is close to 50, so it is not surprising. A random run is rarely exactly the expected number.

7 page 43
Answer: (a) 3/8 (b) 4/11 (c) Bag A
Why: Bag A: 3 red out of 8, so 3/8. Bag B: 4 red out of 11, so 4/11. Compare with a common bottom number, 88: 3/8 = 33/88 and 4/11 = 32/88. 33 is greater than 32, so Bag A gives the better chance.

8 page 44
Answer: (a) 3 (b) 6 (c) 1/12
Why: Count in the table. Total 4 is 1+3, 2+2 and 3+1: 3 totals. Totals less than 5 are 2, 3 and 4: 1 + 2 + 3 = 6 totals. 3 out of 36 is 1/12.

9 page 44
Answer: (a) 2/15 (b) 1/3 (c) 1/6
Why: There are 30 equally likely numbers. Multiples of 7: 7, 14, 21, 28 is 4, so 4/30 = 2/15. Primes: 2, 3, 5, 7, 11, 13, 17, 19, 23, 29 is 10, so 10/30 = 1/3. Squares: 1, 4, 9, 16, 25 is 5, so 5/30 = 1/6.

10 page 44
Answer: (a) 100 (b) 12 (c) Yes
Why: (a) 1/2 × 200 = 100. (b) 112 − 100 = 12. (c) A random run is not exactly 100 each time. 112 is not far from 100, so the coin could still be fair.

11 page 44
Answer: (a) 7 (b) 10 (c) 2
Why: The totals 2 to 12 are not equally likely. 7 happens 6 ways, which is the most. 6 happens 5 ways and 8 happens 5 ways: 10/36. 2 (1+1) and 12 (6+6) happen 1 way each: 2/36.

THINK HARDER

1 page 45
Answer: (a) 18 (b) Yes (c) 5
Why: An even total needs 2 odd dice (3 × 3 = 9 ways) or 2 even dice (9 ways): 18 of 36, so it is fair. In Game 2 Ali wins 6/36 and Ben 30/36. If Ali scores 5 points a win and Ben 1, then 6 × 5 = 30 = 30 × 1: fair.

2 page 45
Answer: (a) 20 (b) 6 (c) 3/10
Why: The first counter has 5 choices and the second has 4: 5 × 4 = 20 equally likely ordered pairs. Both red: 3 choices, then 2: 6 pairs. 6/20 = 3/10.

3 page 45
Answer: (a) 100 (b) 102.8 (c) Yes
Why: (a) 1/6 × 600 = 100. (b) 93 + 108 + 97 + 114 + 102 = 514, and 514 ÷ 5 = 102.8. (c) Every run is near 100 and the mean is very close to 100, so the dice looks fair.

GO FURTHER

1 page 46
Answer: (a) 16 (b) 6 (c) 15
Why: Each flip has 2 results, so 2 × 2 × 2 × 2 = 16 equally likely outcomes. Exactly 2 heads: HHTT, HTHT, HTTH, THHT, THTH, TTHH is 6, so 6/16. At least one tail is every outcome except HHHH: 15/16.

2 page 46
Answer: (a) 1 (b) 100 (c) 50
Why: For any a, only 1 of the 6 values of b matches it: 1 out of 6. So we expect 1/6 × 600 = 100 doubles. The pairs that add to 4 are 1+3, 2+2 and 3+1: 3 of 36, which is 1/12. 1/12 × 600 = 50.

3 page 46
Answer: (a) 100 (b) 1200, 500 (c) 12
Why: A 6 comes up about 1/6 × 600 = 100 times. You pay 600 × 2 = 1200 dollars. With a 5-dollar prize you win about 500, so you lose. Fair means 100 × prize = 1200: the prize is 12 dollars.

CHECK YOURSELF

1 page 47
Answer: 2/5
Why: 15 counters in all and 6 are blue: 6/15. Divide the top and bottom by 3: 2/5.

2 page 47
Answer: 120
Why: Even numbers are 2, 4 and 6, so the probability is 3/6 = 1/2. 1/2 × 240 = 120.

3 page 47
Answer: 1/4
Why: The outcomes are HH, HT, TH and TT, which are equally likely. Only TT has 2 tails: 1/4.

4 page 47
Answer: 100
Why: P(green) = 2/8 = 1/4. 1/4 × 400 = 100.

Unit 7 · Turtle geometry

Your turn page 48
Answer: 40°, 140°
Why: 360° ÷ 9 = 40° for the turn. The inside angle is 180° − 40° = 140°.

START HERE

1 page 49
Answer: Row 1: 72°, 108°. Row 2: 30°, 150°. Row 3: 20 sides, 162°. Row 4: 45 sides, 8°.
Why: Turn = 360 ÷ sides and inside angle = 180 − turn. 5 sides: turn 72, inside 108. 12 sides: turn 30, inside 150. Turn 18: 360 ÷ 18 = 20 sides, inside 162. Inside 172: turn 8, and 360 ÷ 8 = 45 sides.

2 page 49
Answer: (a) B (b) 180°, does not close
Why: Inside 150° means a turn of 180° − 150° = 30°, and 360° ÷ 30° = 12 sides: Program B. Program C also turns 30°, but only 6 × 30 = 180° in all. That is half a turn, so the path does not close.

3 page 49
Answer: (a) 140 (b) No
Why: The total turned grows by 100 each repeat: 100, 200, 300, 400, 500. Take away 360 when the total passes it: 500 − 360 = 140. The turtle faces 140° from its start direction, not 0°, so it is not facing the way it began.

4 page 50
Answer: (a) 55 (b) 125 (c) 540
Why: All the turns add to 360°. The first 4 add to 305, so the last turn is 55° and the inside angle there is 125°. The 5 inside angles add to 5 × 180 − 360 = 540°.

5 page 50
Answer: (a) 20 (b) 160 (c) 2880
Why: Turn: 360 ÷ 18 = 20°. Inside angle: 180 − 20 = 160°. All 18 inside angles: 18 × 160 = 2880°. Check with (18 − 2) × 180 = 16 × 180 = 2880.

6 page 50
Answer: 24°, 30°, 45°, 72°, 120°
Why: The turn must divide 360 exactly. 360 ÷ 24 = 15 sides, ÷ 30 = 12, ÷ 45 = 8, ÷ 72 = 5 and ÷ 120 = 3 sides. 360 ÷ 50, 64, 100 and 150 are not whole numbers.

7 page 50
Answer: (a) 1080 (b) 108 (c) 72
Why: Going round 3 times, the turtle turns 3 × 360 = 1080° in all. 1080 ÷ 10 = 108° at each point. The angle at each point is 180 − 108 = 72°.

8 page 51
Answer: (a) 30 (b) 12 (c) 168
Why: 30 sides need 30 repeats. The turn is 360 ÷ 30 = 12°. The inside angle is 180 − 12 = 168°.

9 page 51
Answer: (a) 360 (b) 300 (c) 360
Why: The rectangle has 4 right-angle turns: 4 × 90 = 360°. The turtle walks 100 + 50 + 100 + 50 = 300 steps. With 80 instead of 50: 100 + 80 + 100 + 80 = 360 steps.

10 page 51
Answer: (a) 360 (b) 270°, Yes (c) 360
Why: Triangle angle 60: 6 × 60 = 360. Square angle 90: 3 × 90 = 270, which is less than 360, so there is a gap of 90. Hexagon angle 120: 2 × 120 + 2 × 60 = 360.

11 page 51
Answer: (a) 6 (b) 12 (c) 18
Why: Turn + inside = 180. If the inside angle is k times the turn, then (k + 1) × turn = 180. k = 2: turn 60, so 6 sides. k = 5: turn 30, so 12 sides. k = 8: turn 20, so 18 sides.

THINK HARDER

1 page 52
Answer: (a) 18 (b) 1800 (c) 5
Why: After k repeats the total turned is 100k. It faces the start direction when 100k is a multiple of 360: 100k = 1800 first, at k = 18 (1800 ÷ 100). 1800 = 5 × 360, so it has turned round 5 times and draws an 18-point star.

2 page 52
Answer: (a) 15 (b) 156 (c) 24
Why: The sum of the inside angles is (n − 2) × 180, so n − 2 = 2340 ÷ 180 = 13 and n = 15. Each inside angle is 2340 ÷ 15 = 156°. The turn is 180 − 156 = 24°. Check: 360 ÷ 24 = 15.

3 page 52
Answer: (a) 120 (b) 6
Why: The angles are 60 + 90 + 90 = 240. 360 − 240 = 120° is left. A regular polygon with inside angle 120° has turn 60°, so 360 ÷ 60 = 6 sides: a hexagon.

GO FURTHER

1 page 53
Answer: (a) −20, 20 (b) −60, 60
Why: Right 10, down 20, left 30, up 40 ends at x = 10 − 30 = −20, y = −20 + 40 = 20. The next 4 moves (50, 60, 70, 80) end at x = −40, y = 40. Each 4 repeats adds −20 to x and 20 to y, so 12 repeats end at (−60, 60).

2 page 53
Answer: (a) 180 (b) 360 (c) 72°, 720°
Why: 5-point star: turn 144, point angle 36, and 5 × 36 = 180. 8-point star going round 3 times: turn 135, point angle 45, and 8 × 45 = 360. 10-point star going round 3 times: turn 108, point angle 72, and 10 × 72 = 720.

3 page 53
Answer: (a) 360 (b) 115
Why: 360 turns of 1° make one full turn, so the path closes. Inside angle 180 − 1 = 179°. The path is 360 steps long. Round ≈ 3.14 × across, so across ≈ 360 ÷ 3.14 = 114.6, about 115.

CHECK YOURSELF

1 page 54
Answer: 9°, 171°
Why: 360 ÷ 40 = 9°. The inside angle is 180 − 9 = 171°.

2 page 54
Answer: 36
Why: 360 ÷ 10 = 36 sides.

3 page 54
Answer: 15
Why: The turn is 180 − 156 = 24°, and 360 ÷ 24 = 15 sides.

4 page 54
Answer: 80°
Why: The turns add to 360. 95 + 80 + 105 = 280, so the last turn is 360 − 280 = 80°.

Unit 8 · Secret codes

Your turn page 55
Answer: VMN
Why: J = 9, A = 0 and B = 1. Adding 12 gives 21, 12 and 13. 21 = V, 12 = M and 13 = N. The code is VMN.

START HERE

1 page 56
Answer: (a) 3 (b) 4 (c) 7 (d) 6
Why: 38 = 5 × 7 + 3, so 3. 100 = 8 × 12 + 4, so 4. 59 = 2 × 26 + 7, so 7. 500 = 19 × 26 + 6, so 6.

2 page 56
Answer: (a) Tuesday (b) Saturday (c) Friday
Why: 40 mod 7 = 5. Thursday and 5 more days: Fri, Sat, Sun, Mon, Tue. 100 mod 7 = 2, so Thursday + 2 is Saturday. 20 mod 7 = 6. Back 6 days from Thursday: Wed, Tue, Mon, Sun, Sat, Fri.

3 page 56
Answer: (a) KHOS (b) RFEJ
Why: HELP: 7, 4, 11, 15 plus 3 is 10, 7, 14, 18, which is KHOS. MAZE: 12, 0, 25, 4 plus 5 is 17, 5, 30, 9. 30 − 26 = 4, so the numbers 17, 5, 4, 9 are RFEJ.

4 page 56
Answer: (a) 6 (b) G (c) 0
Why: X is 23, and 23 + 9 = 32. 32 mod 26 = 6, and 6 is G. For letter 17: 17 + 9 = 26, and 26 mod 26 = 0, which is A.

5 page 57
Answer: (a) WORK (b) LAMP
Why: FXAT: 5, 23, 0, 19 minus 9 is −4, 14, −9, 10. Adding 26 to the negatives gives 22, 14, 17, 10, which is WORK. YNZC: 24, 13, 25, 2 minus 13 is 11, 0, 12, −11. Adding 26 gives 11, 0, 12, 15, which is LAMP.

6 page 57
Answer: (a) 7 (b) QUICK
Why: T = 19 and A = 0, so the shift is 7 (19 + 7 = 26 = 0). Decode XBPJR: 23, 1, 15, 9, 17 minus 7 is 16, −6, 8, 2, 10. −6 + 26 = 20. The numbers 16, 20, 8, 2, 10 are QUICK.

7 page 57
Answer: (a) 2 (b) 9 (c) V
Why: total goes 7, 14, 21, then 28. 28 is more than 25, so 26 is taken away: 2. The 5th is 2 + 7 = 9. The 3rd number, 21, is the letter V.

8 page 58
Answer: (a) 3 (b) 8 (c) 9
Why: (a) 8 + 7 = 15, and 15 − 12 = 3. (b) 5 − 9 = −4, and −4 + 12 = 8. (c) 7 + 50 = 57. 57 − 48 = 9 (4 full turns of 12 are 48), so it is 9 o'clock.

9 page 58
Answer: (a) 7 (b) 19 (c) the same
Why: 15 + 18 = 33, and 33 mod 26 = 7. To undo a shift of 7, go back 7, which is the same as going forward 26 − 7 = 19. A shift of 26 is a full turn round the alphabet, so nothing changes.

10 page 58
Answer: (a) 5 (b) 22 (c) 13
Why: (a) 22 + 7 = 29, and 29 − 24 = 5. (b) 3 − 5 = −2, and −2 + 24 = 22. (c) 100 mod 24 = 4, so 9 + 4 = 13.

11 page 58
Answer: (a) 12 (b) 4
Why: Jumps of 7: 7, 2, 9, 4, 11, 6, 1, 8, 3, 10, 5, 12. All 12 numbers. Jumps of 9: 9, 6, 3, 12. Only 4 numbers, because 9 and 12 share a factor, 3.

THINK HARDER

1 page 59
Answer: (a) 18 (b) 53
Why: Remainder 3 when divided by 5: 3, 8, 13, 18, ... 18 is the first with remainder 4 when divided by 7. The pattern repeats every 35, so the next is 53.

2 page 59
Answer: (a) 90 (b) 6 (c) Monday
Why: 10 Jan to 10 Apr: 31 + 28 + 31 = 90 days. 90 mod 7 = 6 (90 = 12 × 7 + 6). Tuesday and 6 more days is Monday.

3 page 59
Answer: (a) 4 (b) MEET AT NOON
Why: A = 0 became E = 4, so the shift is 4. Decode each letter by taking away 4. QIIX: 16, 8, 8, 23 minus 4 is 12, 4, 4, 19: MEET. EX: AT. RSSR: 17, 18, 18, 17 minus 4 is 13, 14, 14, 13: NOON.

GO FURTHER

1 page 60
Answer: (a) NCWIU (b) CODES (c) the same letter is coded in different ways
Why: MATHS: 12, 0, 19, 7, 18 plus 1, 2, 3, 1, 2 gives 13, 2, 22, 8, 20: NCWIU. DQGFU: 3, 16, 6, 5, 20 minus 1, 2, 3, 1, 2 gives 2, 14, 3, 4, 18: CODES. A letter can be coded in different ways.

2 page 60
Answer: (a) 2 (b) 5
Why: The 1st jump goes from 9 to 14, which is more than 12, so 12 is taken away: 2. After 100 jumps it has moved 500 hours: 9 + 500 = 509. 509 mod 12 = 5 (509 = 42 × 12 + 5), so it ends at 5.

3 page 60
Answer: (a) 7 (b) 2 (c) No
Why: 4 + 7 + 2 = 13, and 20 is the next multiple of 10: check digit 7. 9 + 8 + 1 = 18: check digit 2. 5 + 3 + 6 = 14 needs 6, not 9: not valid.

CHECK YOURSELF

1 page 61
Answer: 1
Why: 9 + 40 = 49. 49 mod 12 = 1, because 4 × 12 = 48. It will be 1 o'clock.

2 page 61
Answer: TUC
Why: N = 13, O = 14, W = 22. Adding 6 gives 19, 20, 28. 28 − 26 = 2. The numbers 19, 20, 2 are T, U, C: TUC.

3 page 61
Answer: ARM
Why: E = 4, V = 21, Q = 16. Taking away 4 gives 0, 17, 12: A, R, M. The word is ARM.

4 page 61
Answer: Sunday
Why: 30 mod 7 = 2. Friday and 2 more days is Sunday.

Unit 9 · Data detectives

Your turn page 62
Answer: 9
Why: The total for 5 weeks is 5 × 10 = 50 dollars. The first 4 weeks add to 8 + 14 + 12 + 7 = 41. Week 5: 50 − 41 = 9 dollars.

START HERE

1 page 63
Answer: (a) 10 (b) 5 (c) 2.75
Why: (a) 40 ÷ 4 = 10. (b) 25 ÷ 5 = 5 (the 0s count as values). (c) 11 ÷ 4 = 2.75.

2 page 63
Answer: (a) 6 (b) 4
Why: Total: 9 + 7 + 5 + 3 = 24 blocks, shared over 4 towers: 24 ÷ 4 = 6. Tower A gives away 3 blocks and tower B gives away 1: 4 blocks move. Towers C and D receive them (1 and 3).

3 page 63
Answer: (a) 30 (b) 10
Why: Total mass = mean × how many = 6 × 5 = 30 kg. The 3 known parcels weigh 4 + 9 + 7 = 20 kg, so the other 2 weigh 30 − 20 = 10 kg together.

4 page 63
Answer: (a) 75 (b) 82
Why: Total of 4 scores = 4 × 82 = 328. The known scores add to 78 + 90 + 85 = 253. The 4th score is 328 − 253 = 75. A 5th score equal to the mean, 82, keeps the mean at 82.

5 page 64
Answer: (a) stays the same (b) 14 (c) 11
Why: (a) A new value equal to the mean does not change it: the mean stays at 12. (b) 10 + 12 + 14 + 20 = 56, and 56 ÷ 4 = 14. (c) 10 + 12 = 22, and 22 ÷ 2 = 11.

6 page 64
Answer: (a) 7 (b) 7 (c) 8 (d) 5
Why: 5, 7, 7, 9, 12: median 7, mode 7, mean 40 ÷ 5 = 8. 2, 4, 6, 8: the middle two are 4 and 6, so the median is 5.

7 page 64
Answer: (a) 2800 (b) 1700 (c) median
Why: Total 19600, and 19600 ÷ 7 = 2800. In order the middle (4th) value is 1700. The owner's 9500 pulls the mean up, so the median is more typical.

8 page 64
Answer: (a) 200 (b) 252 (c) 42
Why: Total of 5 children = 5 × 40 = 200 kg. With the 6th child: 200 + 52 = 252 kg. New mean = 252 ÷ 6 = 42 kg.

9 page 65
Answer: (a) 72 (b) 12 (c) 2
Why: Total: 14 + 9 + 17 + 12 + 8 + 12 = 72. Mean = 72 ÷ 6 = 12. Above 12: 14 and 17 is 2 pupils. (Two pupils have exactly 12.)

10 page 65
Answer: (a) 24 (b) 6 (c) 8
Why: i goes 1, 2, 3, 4 and the loop adds 7, 9, 3 and 5: total 24. The last line says 24 ÷ 4 = 6. If it divided by 3 it would say 24 ÷ 3 = 8.

11 page 65
Answer: (a) 13 (b) 16
Why: Add 5 to each: 9, 13, 17. 39 ÷ 3 = 13, which is the old mean + 5. Double each: 8, 16, 24. 48 ÷ 3 = 16, which is the old mean × 2.

THINK HARDER

1 page 66
Answer: (a) 40 (b) 5 (c) 11
Why: The ages add up to 5 × 8 = 40. In order the ages are a, b, 7, d, 12. The mode 5 appears at least twice, and only a and b are below 7, so a = b = 5. Then 5 + 5 + 7 + d + 12 = 40, so d = 40 − 29 = 11. The ages are 5, 5, 7, 11, 12.

2 page 66
Answer: (a) 480 (b) 810 (c) 43 (d) boys
Why: Girls: 12 × 40 = 480 kg. Boys: 18 × 45 = 810 kg. All 30: 480 + 810 = 1290 kg. 1290 ÷ 30 = 43. The mean of the whole class is 43 kg, not 42.5, because there are more boys.

3 page 66
Answer: (a) 13.8 (b) 4.5
Why: The total time is 3 × 4.6 = 13.8 minutes. The known laps add to 3.9 + 5.4 = 9.3. The third lap took 13.8 − 9.3 = 4.5 minutes.

GO FURTHER

1 page 67
Answer: (a) 109 (b) 10 (c) 54.5
Why: The 1st and last add to 7 + 102 = 109. So do the 2nd and the 2nd last (12 + 97). 20 numbers make 10 pairs of 109, so the total is 1090. The mean is 1090 ÷ 20 = 54.5, which is halfway between 7 and 102.

2 page 67
Answer: (a) 7 (b) 5.6 (c) 8
Why: Total 4 + 8 + 6 + 10 + 7 = 35 and 35 ÷ 5 = 7. With repeat 4 the total is only 28 and 28 ÷ 5 = 5.6. With 13 added the total is 48 and 48 ÷ 6 = 8.

3 page 67
Answer: (a) 13 (b) 1.3 (c) 1
Why: Total goals: 0 × 2 + 1 × 4 + 2 × 3 + 3 × 1 = 13. Mean = 13 ÷ 10 = 1.3. In order, the 5th and 6th of the 10 values are both 1, so the median is 1.

CHECK YOURSELF

1 page 68
Answer: 8
Why: 6 + 10 + 2 + 14 = 32, and 32 ÷ 4 = 8.

2 page 68
Answer: 6, 4
Why: In order: 4, 4, 6, 9, 11. The middle value is 6, and 4 appears most often.

3 page 68
Answer: 7
Why: Total = 5 × 6 = 30. 30 − 3 − 7 − 4 − 9 = 7.

4 page 68
Answer: 72
Why: The 4 tests total 4 × 70 = 280. With the 5th: 280 + 80 = 360, and 360 ÷ 5 = 72.

Unit 10 · Showcase challenge

Your turn page 69
Answer: Saturday
Why: 19 mod 7 = 5, because 19 = 2 × 7 + 5. pos = 5 + 1 = 6, and item 6 of days is Saturday.

START HERE

1 page 70
Answer: (a) 7, 3 (b) 353 (c) 23
Why: The difference is 7. Each term is 3 more than 7, 14, 21, 28, so the rule is 7n + 3. The 50th term is 7 × 50 + 3 = 353. 7n + 3 = 164 gives 7n = 161, so n = 23.

2 page 70
Answer: (a) 97, 101, 103, 107, 109 (b) 2 × 3 × 5 × 11
Why: 91, 93, 95, 99, 105 and 111 are not prime (7 × 13, 3 × 31, 5 × 19, 9 × 11, 5 × 21, 3 × 37). The primes are 97, 101, 103, 107, 109. 330 = 3 × 11 × 2 × 5.

3 page 70
Answer: (a) −11 (b) 27
Why: −6 + 15 = 9. Then 9 − 20 = −11. From −6 up to 21 is 6 steps to 0 and 21 more: 27.

4 page 70

Answer: (a) 10 (b) 13
Why: (a) OUT 43: 43 + 7 = 50, and 50 ÷ 5 = 10. (b) 6n + 1 = 79 gives 6n = 78 and n = 13.

5 page 70

Answer: (a) 5/8 (b) 200 (c) Yes
Why: (a) 8 counters, 5 red: 5/8. (b) 1/6 × 1200 = 200. (c) An even number is 2, 4 or 6, and 1, 3 or 5 is the other 3 numbers: 3 out of 6 each, so the game is fair.

6 page 71

Answer: (a) 30 (b) 36 (c) 24
Why: (a) 3 × 5 × 2 = 30. (b) 9 friends: 9 × 8 ÷ 2 = 36. (c) 4 choices, then 3, then 2: 4 × 3 × 2 = 24.

7 page 71

Answer: (a) 7 (b) VJCQ
Why: (a) 4 + 75 = 79. 79 mod 12 = 7, because 6 × 12 = 72. (b) M = 12, A = 0, T = 19, H = 7. Add 9: 21, 9, 28, 16. 28 − 26 = 2. The letters are V, J, C, Q.

8 page 71

Answer: (a) 16 (b) 10.5
Why: (a) The total of 4 numbers is 4 × 15 = 60. 11 + 19 + 14 = 44, so the 4th is 16. (b) In order: 4, 9, 12, 15. The middle two are 9 and 12, so the median is halfway between them: 10.5.

9 page 71

Answer: (a) 34 (b) 4 (c) 50
Why: Each round takes away 8: 42, 34, 26, 18. The score 18 is below 20, so it resets to 50 after round 4. Rounds 5 to 8 repeat the pattern: 42, 34, 26, 18, and the score resets to 50 again after round 8. After round 6 the score is 34.

10 page 72

Answer: (a) 90 (b) 176
Why: 360 ÷ 4 = 90 sides. The inside angle is 180 − 4 = 176°.

11 page 72

Answer: D, B, C, A, E
Why: You need an idea (D) before you can do the maths (B). You build the blocks (C) after the maths, test them (A) once they exist, and add a twist (E) when the basic version works.

12 page 72

Answer: (a) 161 (b) 26 (c) 6, 5
Why: n goes 11, 17, 23, ... and after 25 jumps it is 11 + 25 × 6 = 161. From the 1st term to the 26th term is 25 jumps, so 161 is the 26th term. The rule is 6n + 5 (6 × 26 + 5 = 161).

13 page 72

Answer: (a) 4 (b) 1/4 (c) 90
Why: A product of 12 comes from 2 and 6, 3 and 4, 4 and 3, 6 and 2: 4 results. The product is odd only when both numbers are odd: 3 × 3 = 9 results, 9/36 = 1/4. 1/4 × 360 = 90.

14 page 72

Answer: (a) −13 (b) 4
Why: 7 moves of −4 make −28. 15 − 28 = −13. After 4 repeats x = 15 − 16 = −1, which is the first time below 0 (after 3 repeats x = 3).

THINK HARDER

1 page 73

Answer: (a) 9 (b) 35 (c) Yes
Why: 5, 11, 17, 23, 29, 35, 41, 47, 53, 59. 35 = 5 × 7 is the only one that is not prime, so 9 of the 10 are prime. 6n is even and an even number take away 1 is odd, so every term is odd.

2 page 73

Answer: (a) 370 (b) 456 (c) 86
Why: The 5 tests total 5 × 74 = 370. The 6 tests total 6 × 76 = 456. The 6th score is 456 − 370 = 86.

3 page 73

Answer: (a) 19, 37, 73 (b) 37 and 73
Why: Digits adding to 10: 19, 28, 37, 46, 55, 64, 73, 82, 91. Primes: 19, 37, 73. Reversed: 91 = 7 × 13 is not prime, but 73 and 37 both are.

GO FURTHER

1 page 74

Answer: (a) 3, 1 (b) 61 (c) 16
Why: From 2 km to 5 km is 3 km more, and the fare rises by 16 − 7 = 9 dollars, so each km costs 9 ÷ 3 = 3 dollars. Then 7 = 3 × 2 + 1, so there is a 1 dollar start fee. The rule is n × 3 + 1. A 20 km ride costs 3 × 20 + 1 = 61 dollars. A fare of 49 dollars means 3n = 48, so n = 16. (For your twist, start with a ride you can check by hand.)

2 page 74

Answer: (a) 2 (b) 200 (c) 400
Why: The numbers greater than 6 are 7 and 8, which is 2 results out of 8, so 1/4. 1/4 × 800 = 200. The numbers greater than 4 are 5, 6, 7 and 8, which is 4 out of 8, so the count would be 1/2 × 800 = 400. A real run is near these numbers, not exactly equal.

3 page 74

Answer: (a) 8 (b) 16 (c) 52
Why: n = 6: 6, 3, 10, 5, 16, 8, 4, 2, 1. That is 8 steps (halve if even, or 3n + 1 if odd). n = 7: 7, 22, 11, 34, 17, 52, 26, 13, 40, 20, 10, 5, 16, 8, 4, 2, 1 is 16 steps. The largest number reached from 7 is 52.

CHECK YOURSELF

1 page 75

Answer: 7, 3
Why: The difference is 7. Each term is 3 less than 7, 14, 21, 28, so the rule is 7n − 3.

2 page 75

Answer: −4
Why: −7 + 12 = 5. Then 5 − 9 = −4.

3 page 75

Answer: 7/15
Why: 7 red out of 15 counters is 7/15, which is already in simplest form.

4 page 75

Answer: 7
Why: 3 + 100 = 103. 103 mod 12 = 7, because 8 × 12 = 96. It will be 7 o'clock.